

Observational Data and Nonresponse

STA304: Surveys, Sampling, and Observational Data

Emily Somerset, Tuesday June 9, 2026

Housekeeping

- Two more lectures left. Lecture 9 (today, Tuesday) and Lecture 10 (Thursday).
- Final Exam on Tuesday, June 23rd.
- Course evaluation form 
- Final Exam page + formula sheet + practice problems will be posted this Friday.
- Learning Activity 9 due Friday June 12 at 12pm.
- Learning Activity 10 due Tuesday June 16 at 11:59pm.



Survey Sam & Obs Data
STA304H1-F-LEC0101

Students

https://go.blueja.io/mZZ6LU4YWU6SOS5_tYY4tQ

Learning Outcomes

- ☐ Distinguish between observational studies and randomized experiments and explain the types of research questions each can answer.
- ☐ Explain the concept of a causal effect and describe why randomized experiments can be used to investigate causal relationships
- ☐ Describe the missing data mechanisms MCAR, MAR, and NMAR, and explain how they differ.
- ☐ Evaluate the potential impact of nonresponse and missing data on survey estimates.
- ☐ Describe common strategies for reducing nonresponse in surveys.
- ☐ Explain the rationale behind the two-stage approach to nonresponse adjustment and describe how it can reduce nonresponse bias.
- ☐ Construct estimators under a two-stage nonresponse design

7 Types of Studies

Introduction

- In this course, we've primarily dealt with sampling designs and inference methods associated with them.
- After we have conducted the survey, and have collected the data, we have observational data

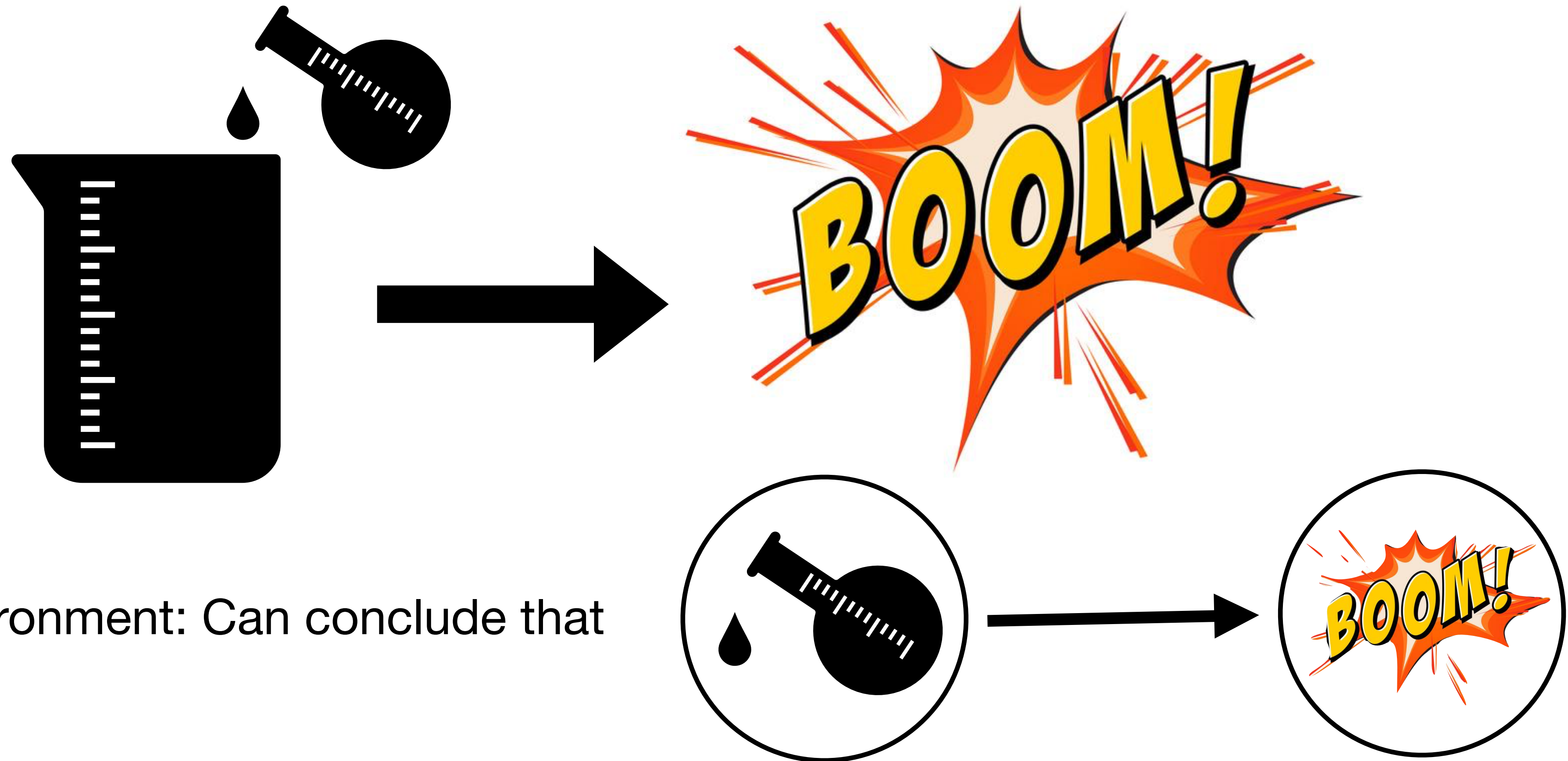
7 Types of Studies

Introduction

- We will first discuss the difference between observational studies and randomized experiments.
- This will help you understand what kind of **research questions you can answer** using randomized experiments vs observational studies.

7.1 Randomized Experiments

Causal effects



Controlled environment: Can conclude that

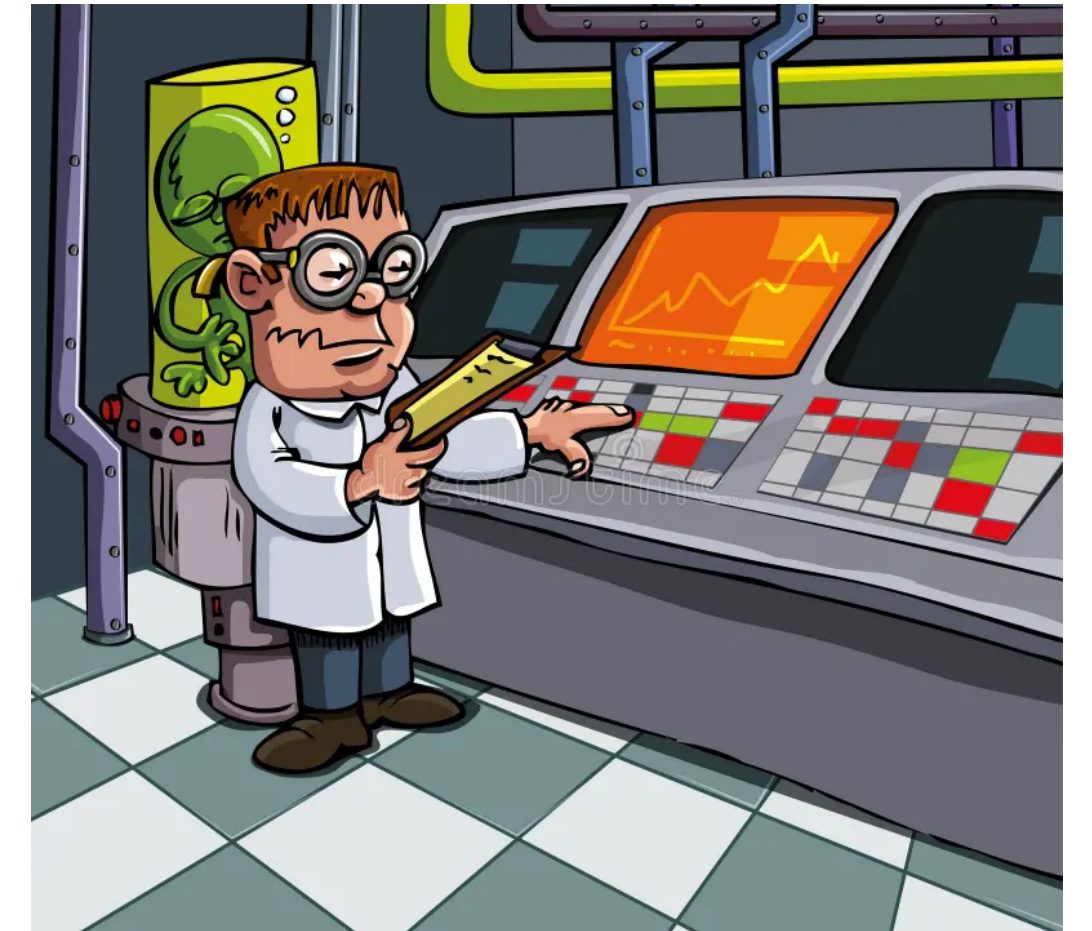
The arrow is a **causal relationship or effect**.

7.1 Randomized Experiments

Causal effects

Another example is a scientist behind a control panel:

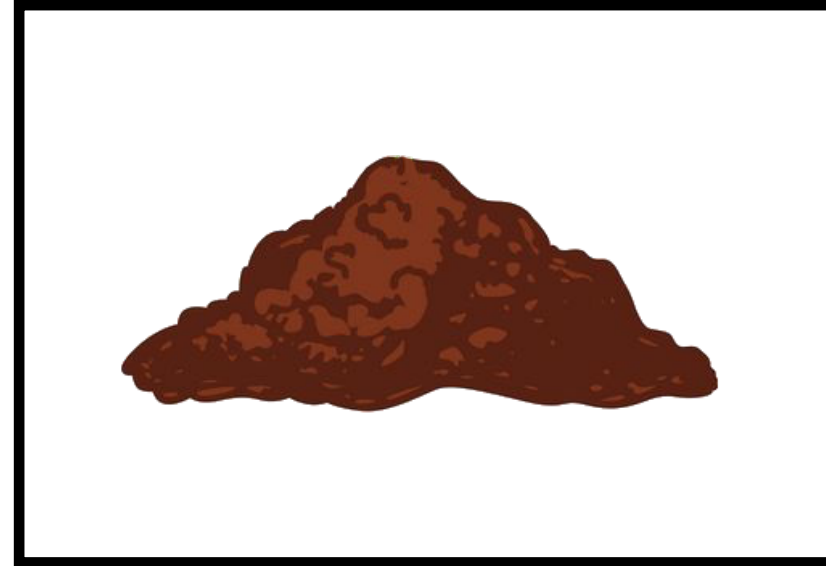
Manipulating temperature



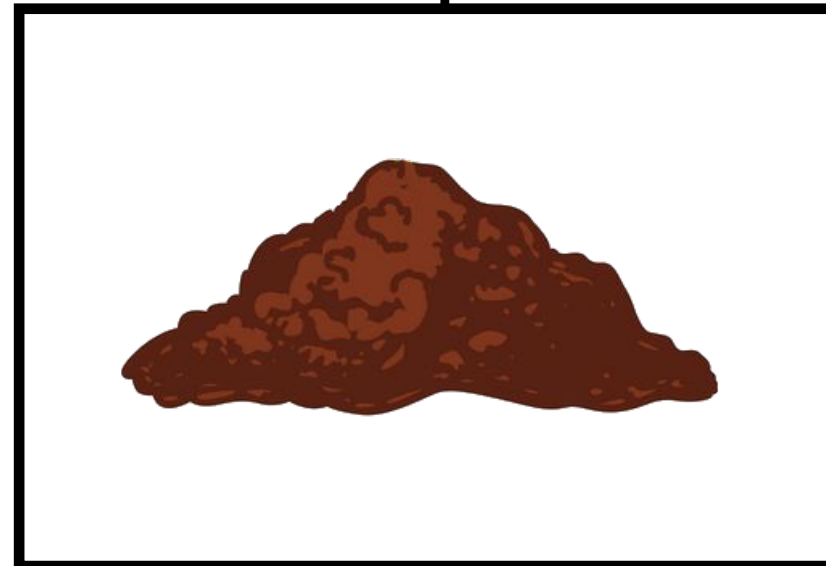
7.1 Randomized Experiments

Causal effects

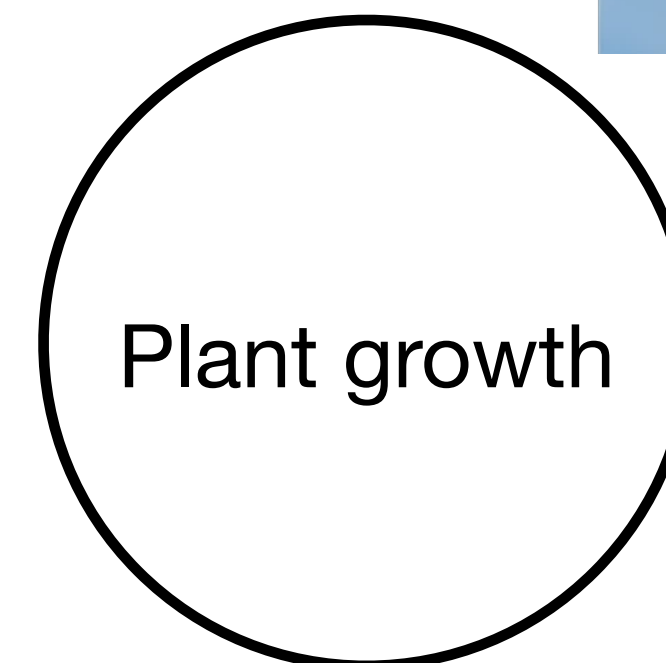
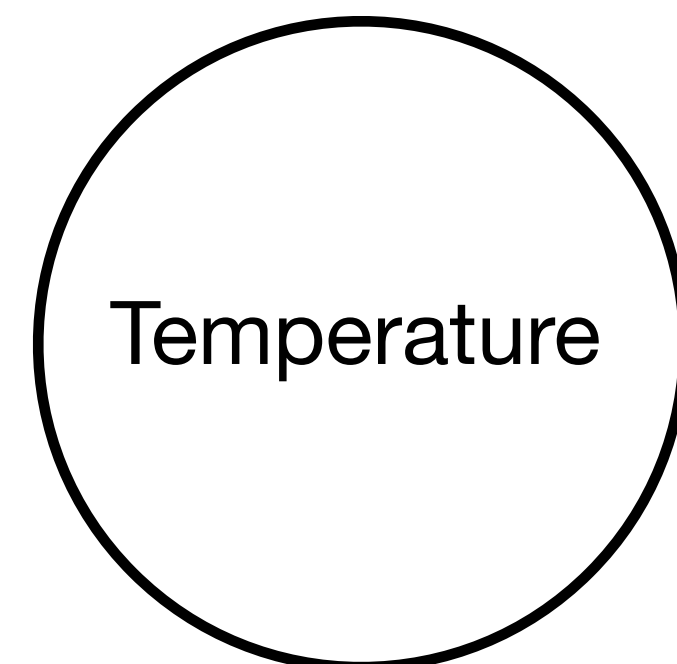
High temperature



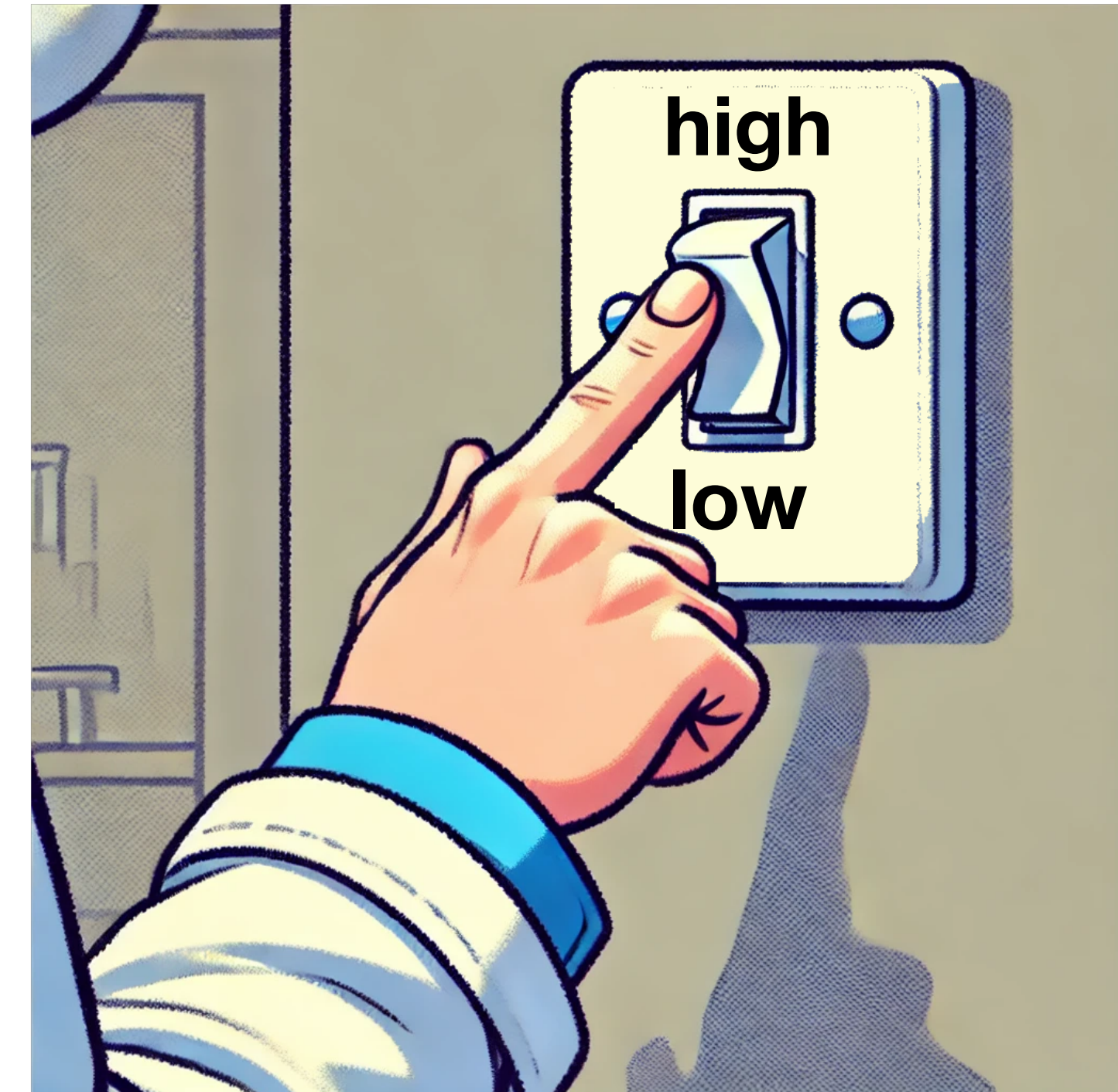
Low temperature



Conclude temperature has a **causal effect** or **relationship** on plant growth or:



Manipulating temperature



7.1 Randomized Experiments

Causal effects

Lots of other questions we can conduct experiments to answer:

- Is a new drug an effective cure for a disease?
- Does exposure to daylight improve sleep quality?
- Does the use of manure increase plant growth?
- Which combination of protein and carbohydrate sources provide the best nutrition for growing kittens?



7.1 Randomized Experiments

Causal effects

Example: Is a new drug an effective cure for a disease?

Ideally we would like to compare a disease response on the same person using the **new drug** and the **placebo**.

However, we can't do this!

Example: Once a person takes a placebo and is cured, you can't go back in time and have them take the drug.

Manipulate treatment

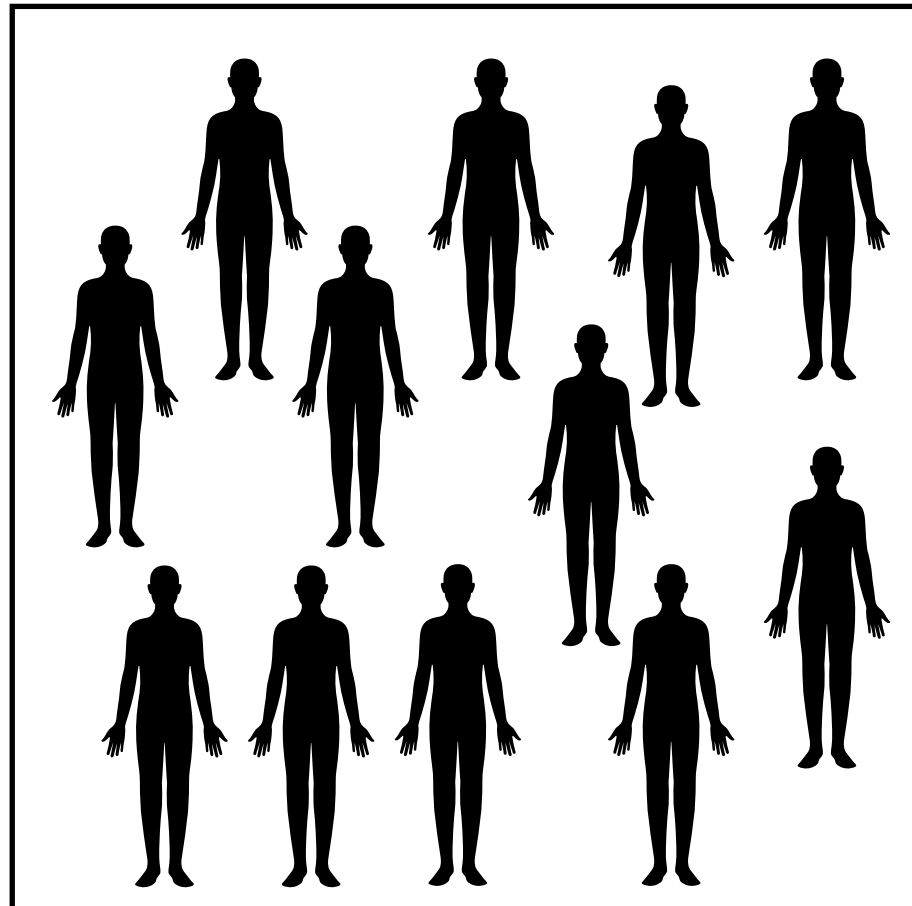


7.1 Randomized Experiments

Causal effects

Instead we place people into a **new drug** group and a **placebo** group and measure responses within each group.

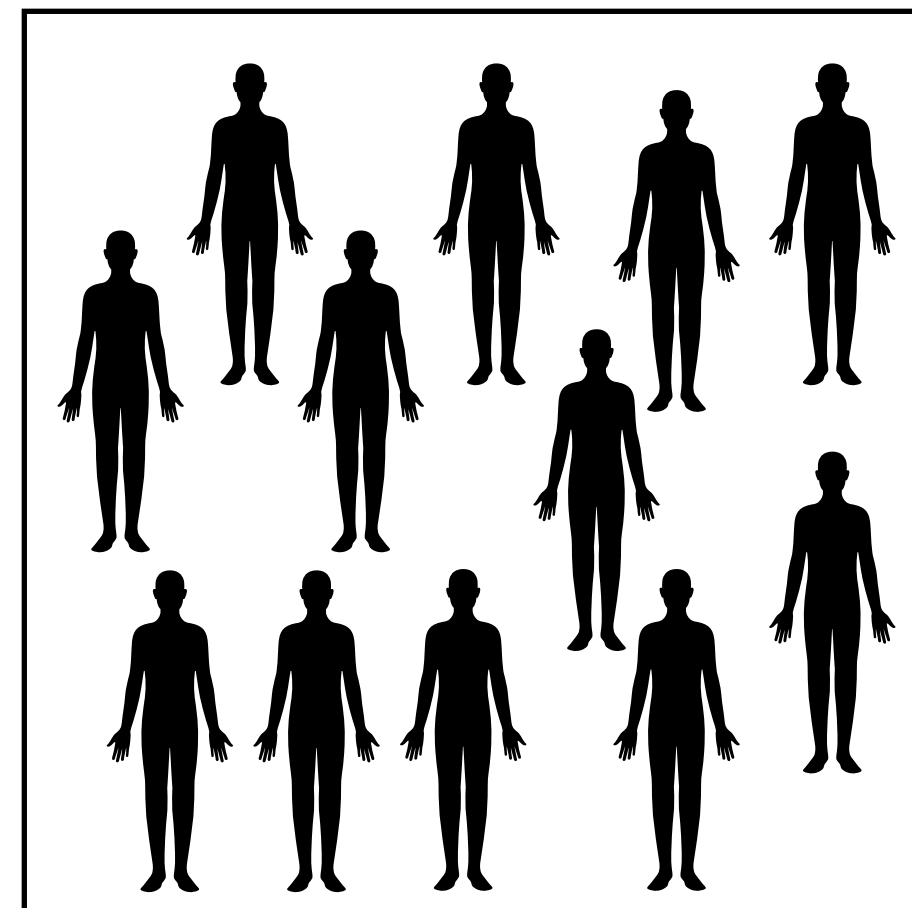
A) new drug



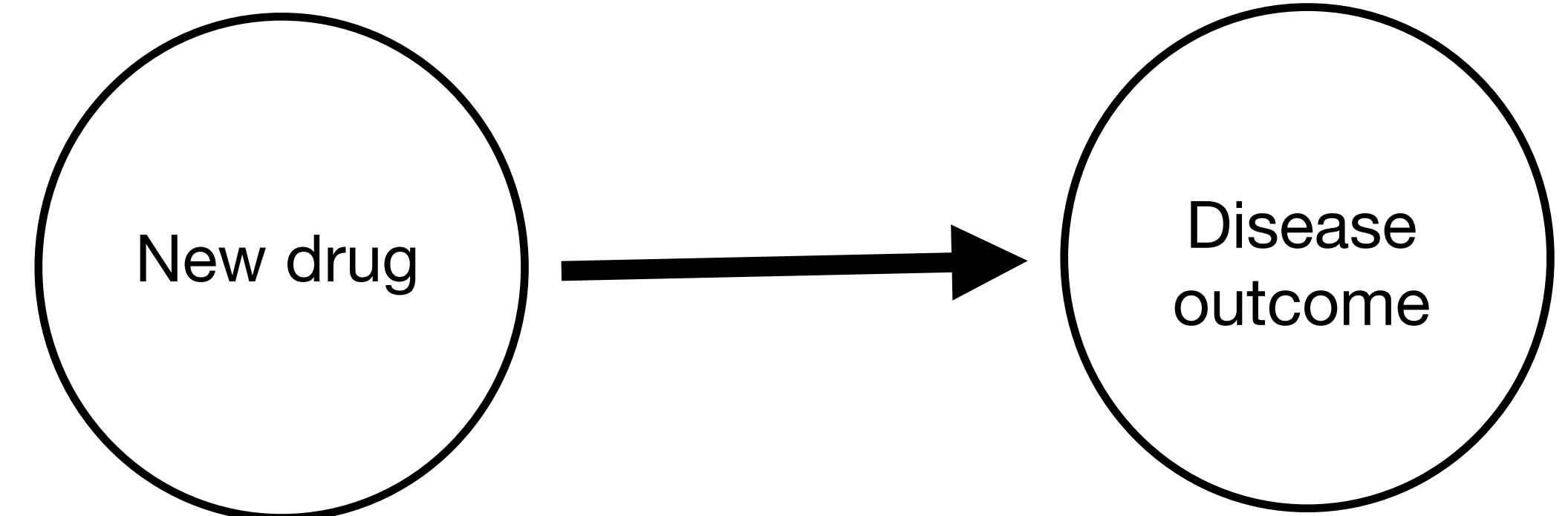
70% cured

Under ideal conditions (which we will talk about later), we can conclude:

B) placebo



8% cured



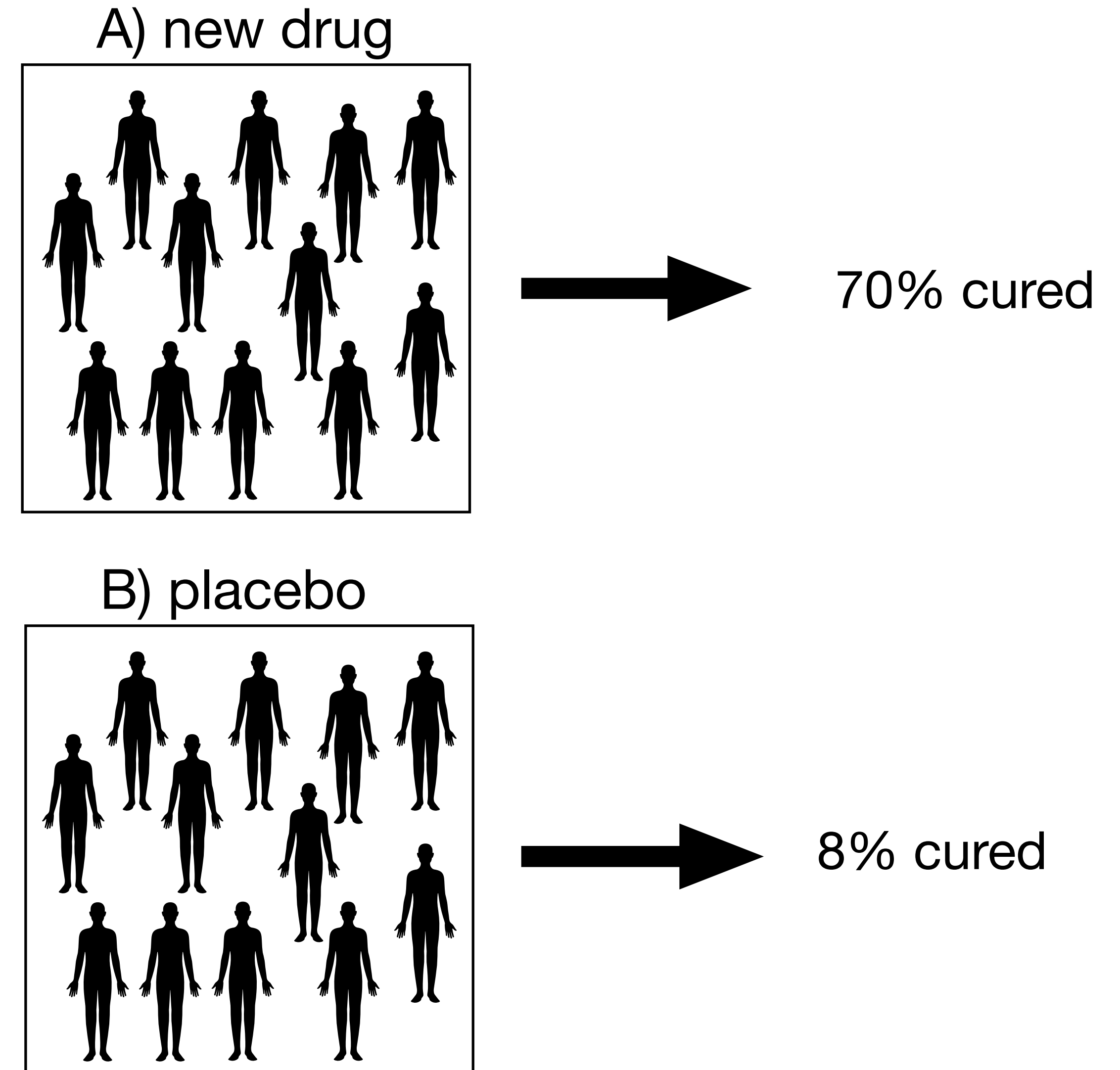
New drug causes increase in probability of disease cure

7.1 Randomized Experiments

Causal effects

In our design of this experiment:

- **Treatment groups:** A) new drug,
B) placebo
- **Response:** Disease status of a patient



7.1 Randomized Experiments

Causal effects

Let’s design an experiment together! What are possible treatment groups and responses to answer the question: **Does the use of manure increase plant growth?**



- Treatments:

- Response:

Plot A	Plot F	Plot K	Plot P
Plot B	Plot G	Plot L	Plot Q
Plot C	Plot H	Plot M	Plot R
Plot D	Plot I	Plot N	Plot S
Plot E	Plot J	Plot O	Plot T

7.1 Randomized Experiments

Experimental Design

Components of experimental design

A. Treatments: Different procedures/treatments we want to compare

B. Experimental unit: Unit to which we want to assign to a treatment

Example: Patient (we are assigning a patient to a drug or placebo),

Plot of land (we are assigning a plot of land to a manure treatment)

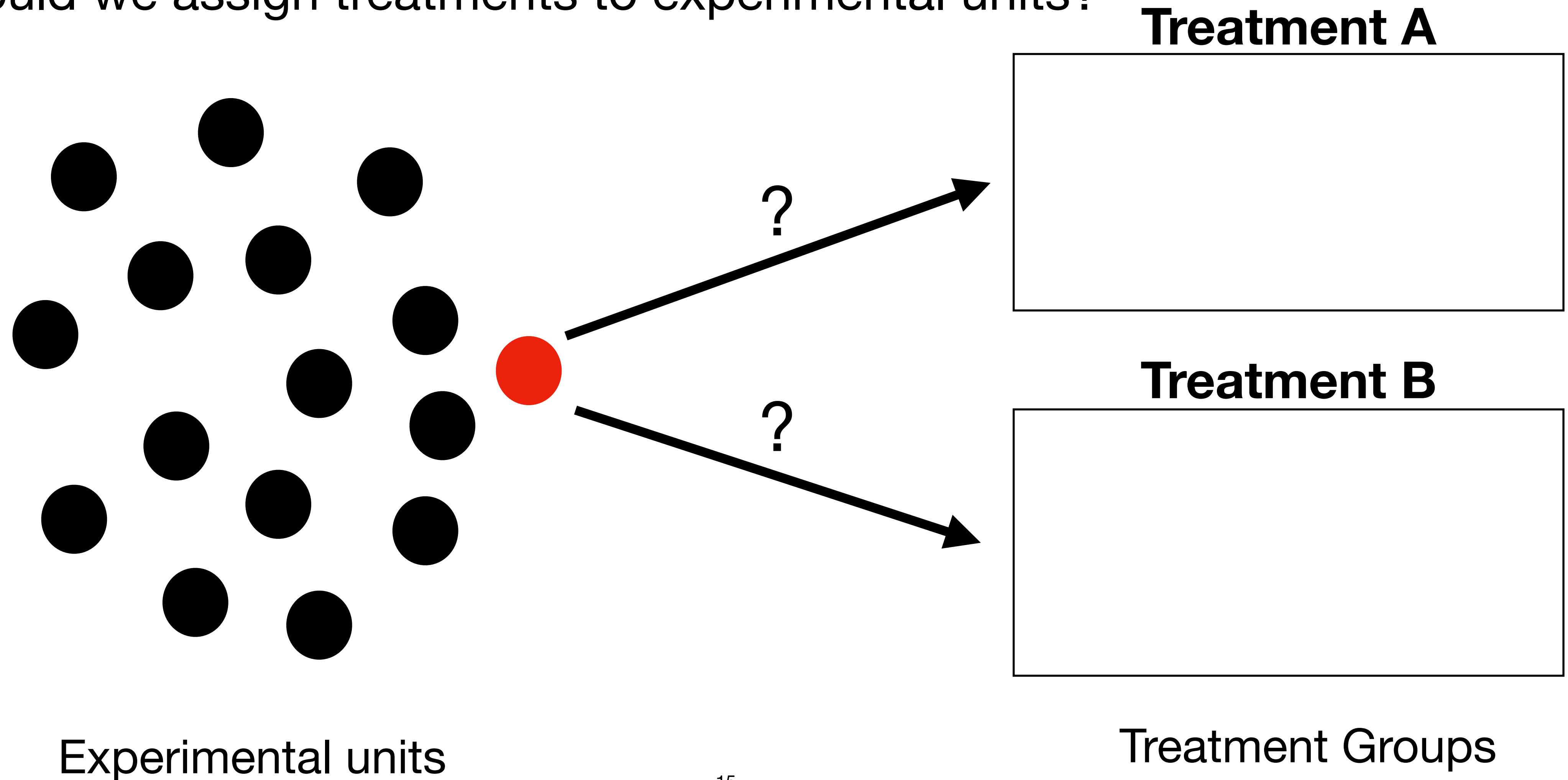
C. Response(s): Outcome(s) we observe after applying a treatment to an experimental unit

D. Method to assign treatments: The method we will use to assign an experimental unit to a treatment group.

7.1 Randomized Experiments

Experimental Design

How should we assign treatments to experimental units?



7.1 Randomized Experiments

Experimental Design

Consider the following question:

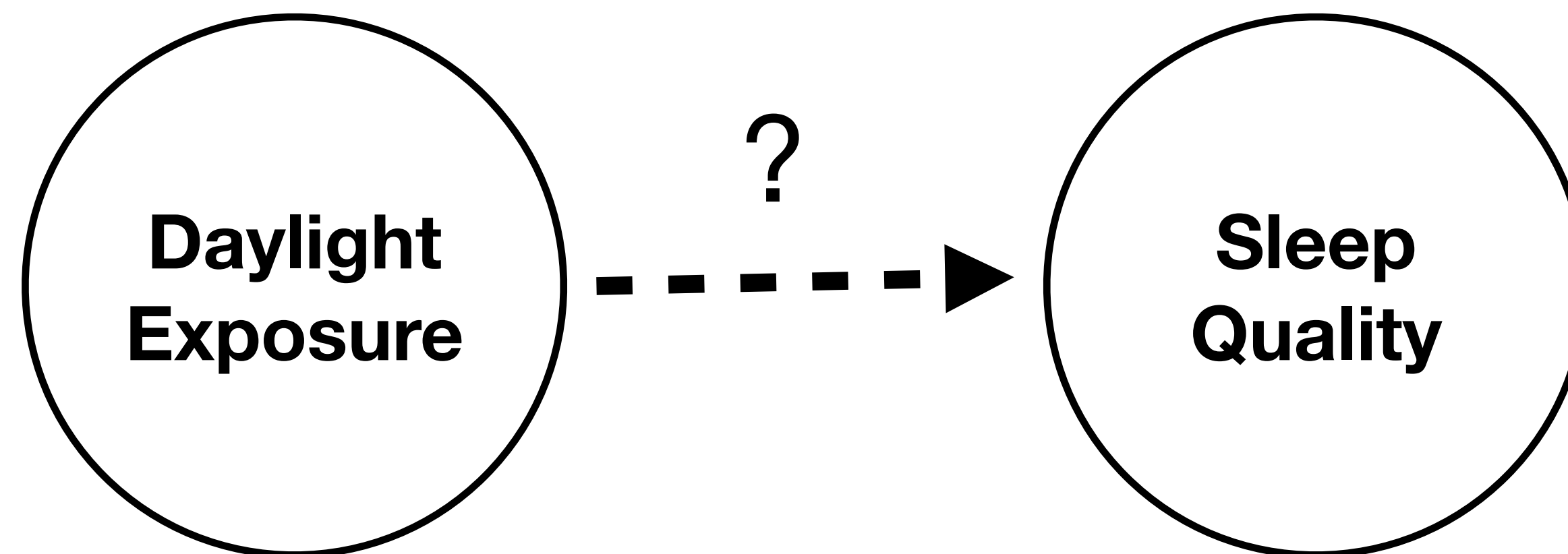
Does exposure to daylight affect sleep quality?



7.1 Randomized Experiments

Experimental Design

Goal: Conduct an experiment to measure the **causal effect of daylight exposure on sleep quality**:

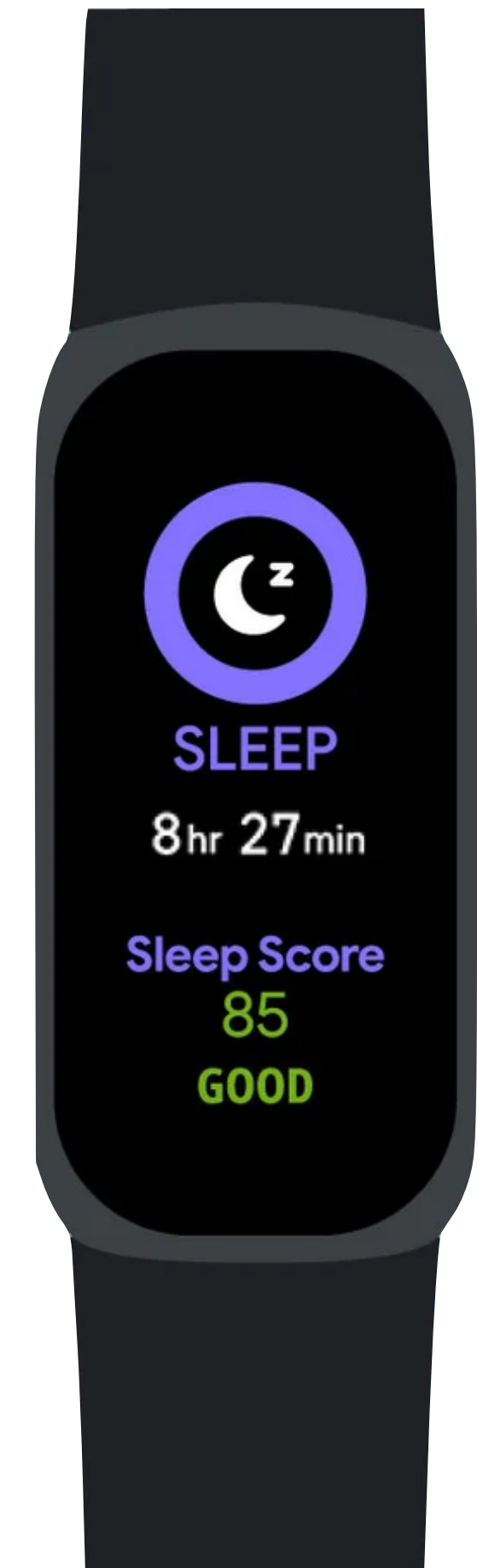


7.1 Randomized Experiments

Experimental Design

Design of this experiment:

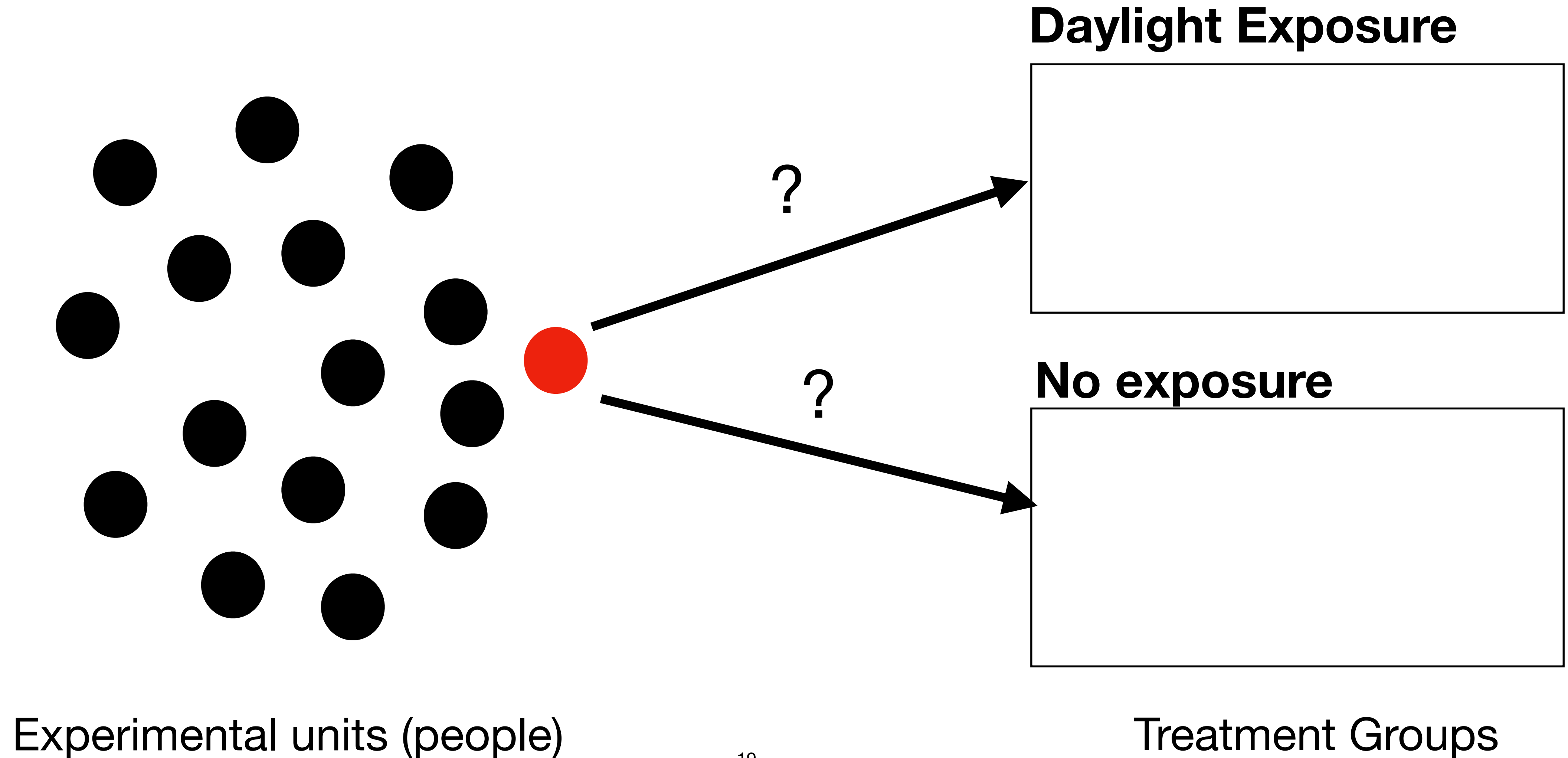
- **Treatment groups:**
 - A) At least 1 hour of daylight exposure
 - B) No exposure
- **Experimental unit:** Person in our pool of volunteers
- **Response:** Sleep score from a Fitbit Charge 6



7.1 Randomized Experiments

Experimental Design

How should we assign treatments to the experimental units?



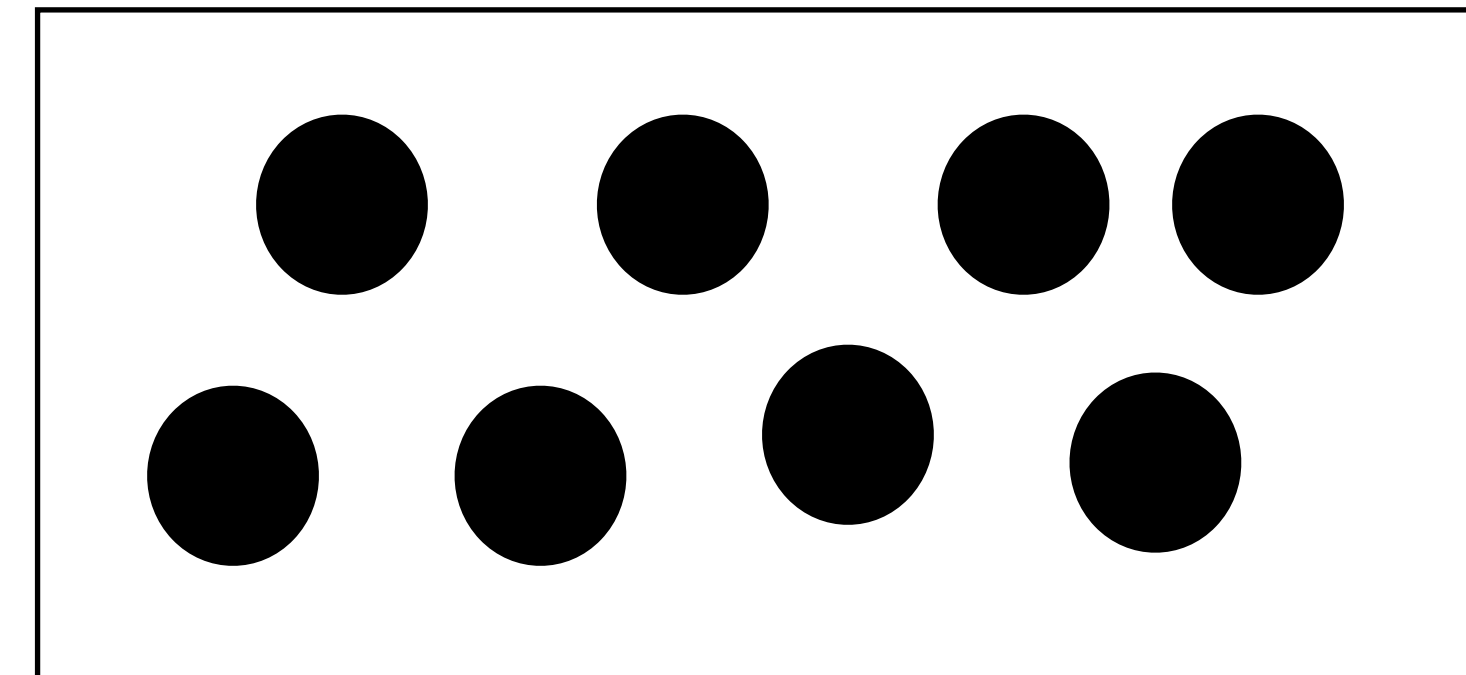
7.1 Randomized Experiments

Confounders

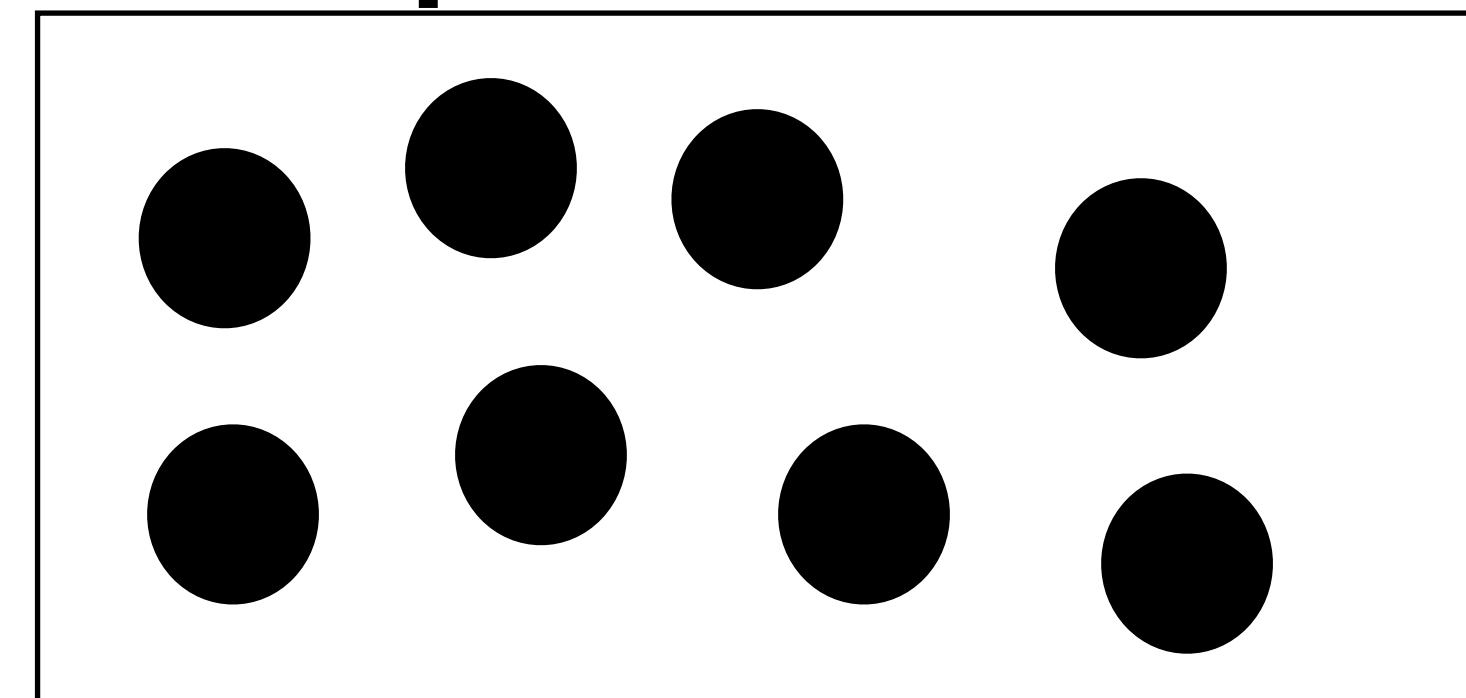
Example 1: We let each person decide what treatment group they want to be in.

How might the people in the **exposure** group be different from the **no exposure** group?

Daylight Exposure



No exposure

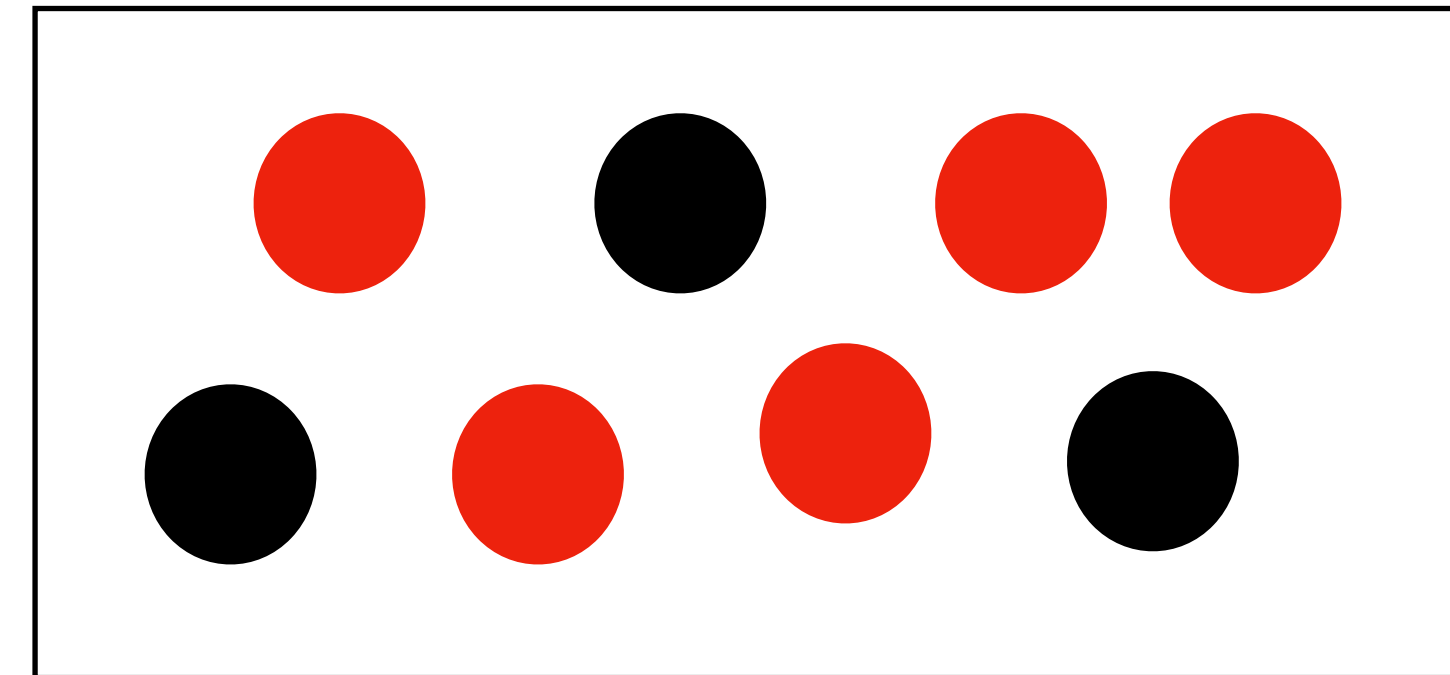


7.1 Randomized Experiments

Confounders

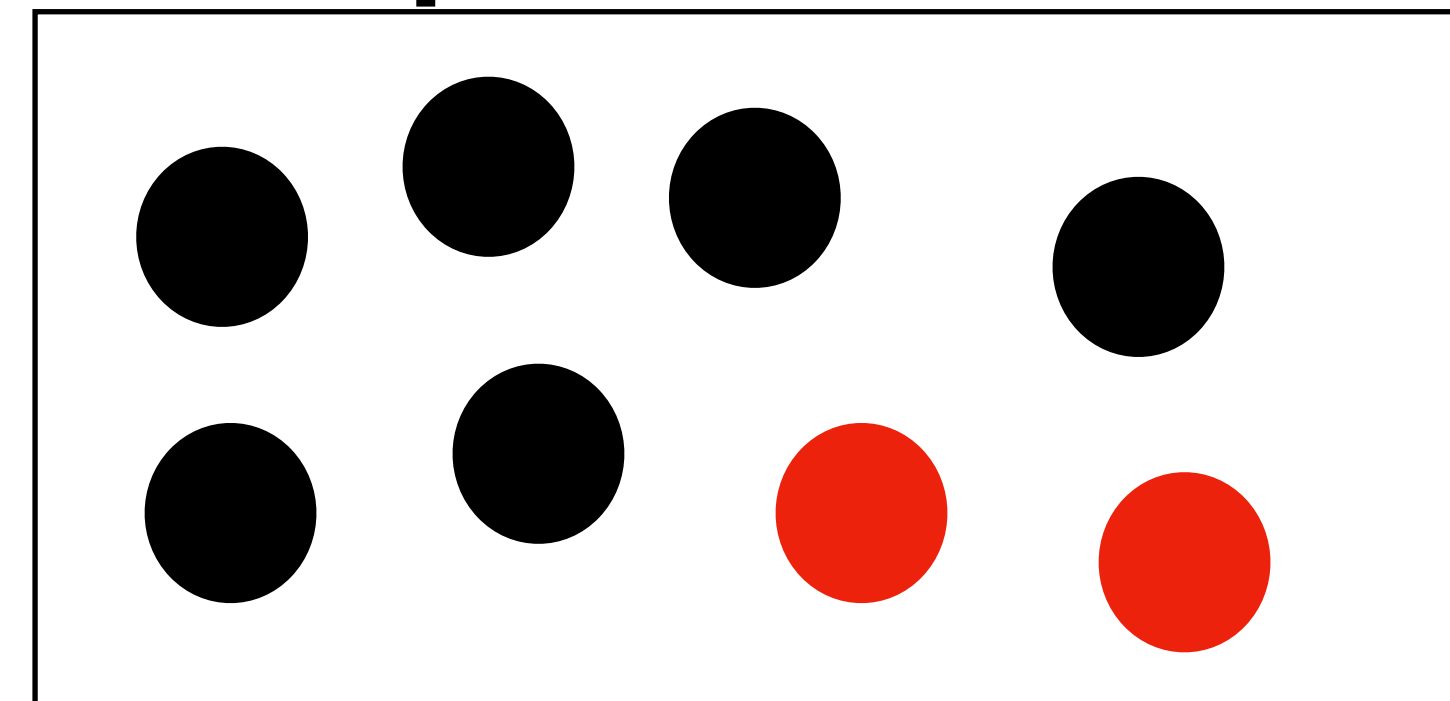
People who are **physically active** (e.g play soccer, football, walk to gym) might self-select into daylight exposure group.

Daylight Exposure



62.5%
physically
active

No exposure

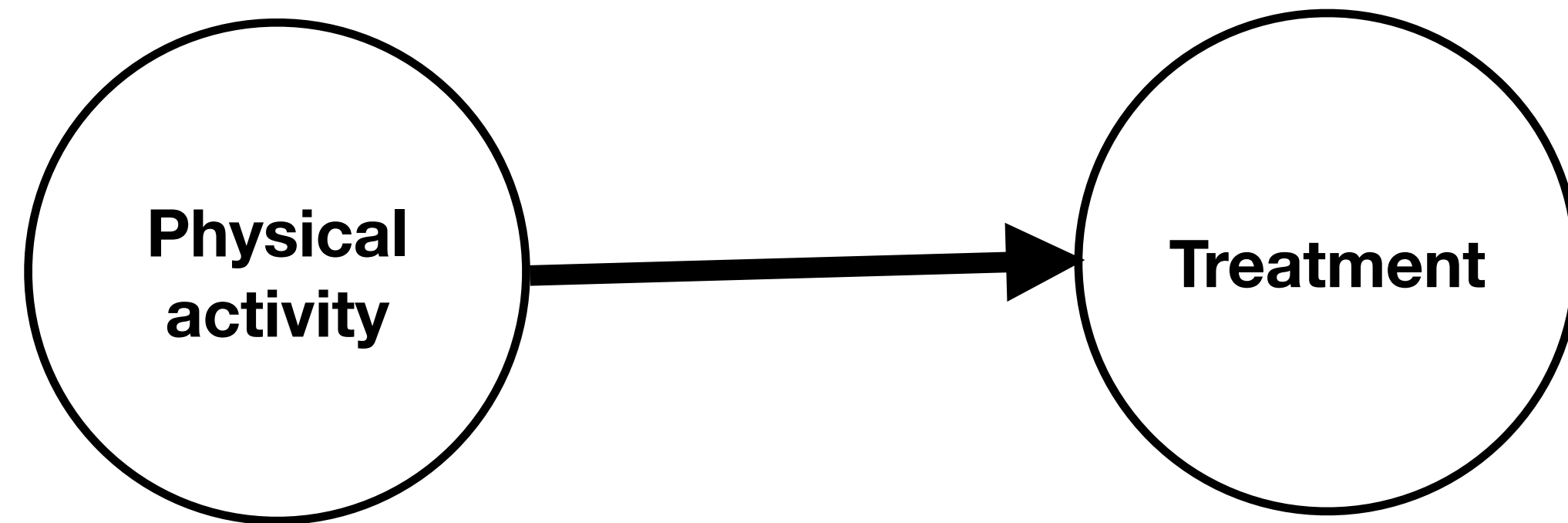


25%
physically
active

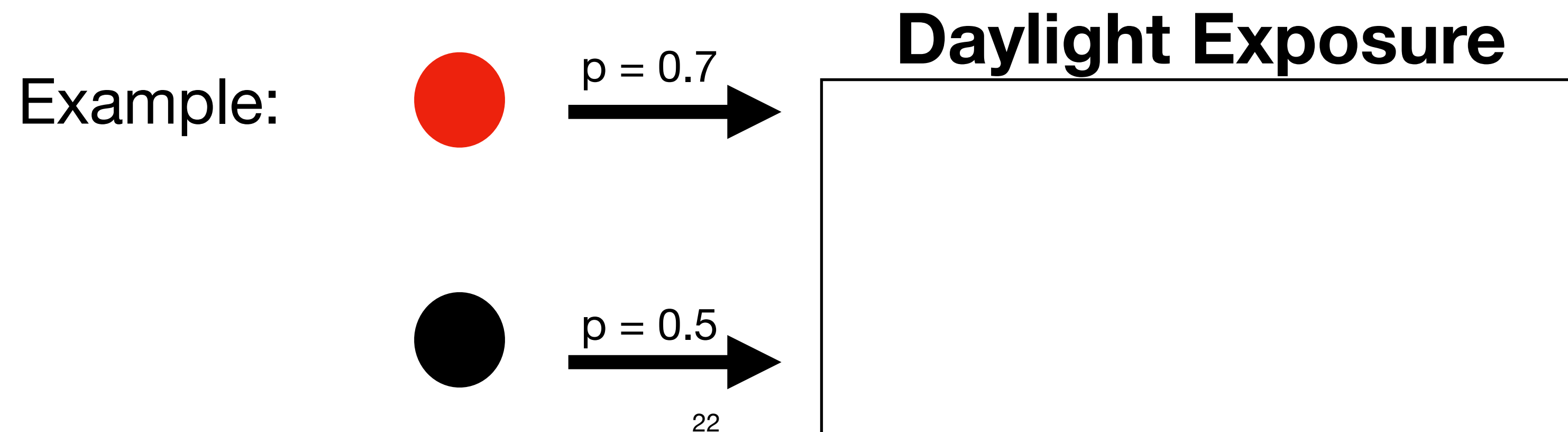
7.1 Randomized Experiments

Confounders

Using this assignment method we've created a causal link between:



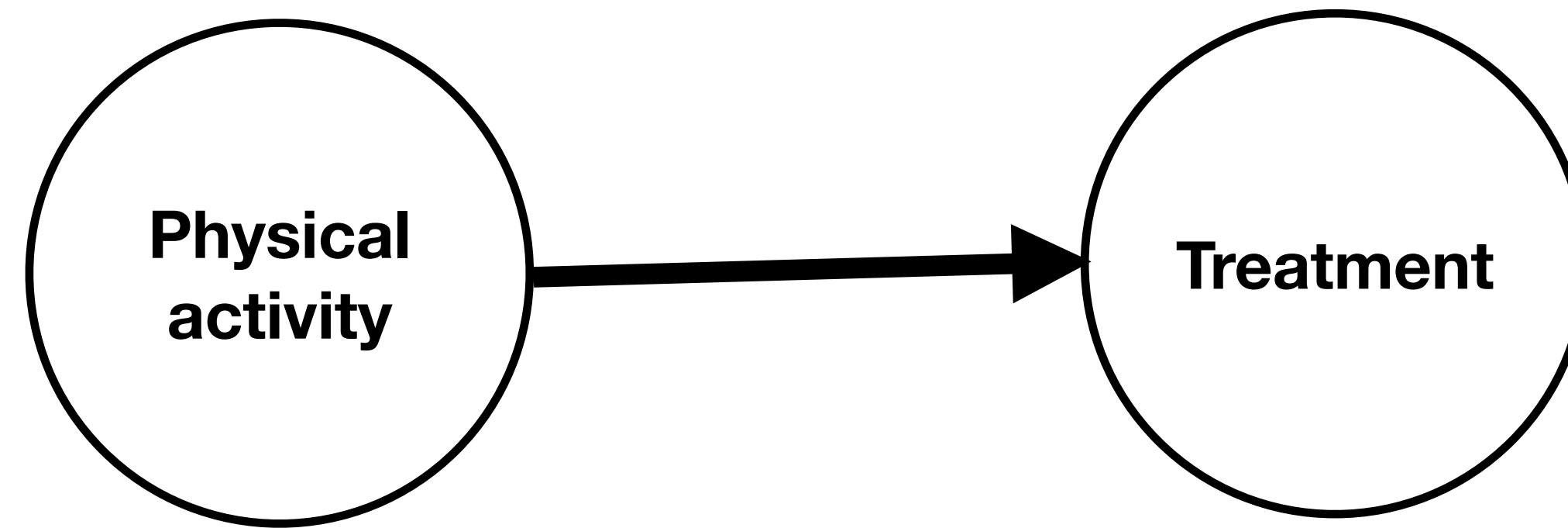
Physical activity causes an increase in probability that a person will self-select into the daylight exposure group



7.1 Randomized Experiments

Confounders

Using this assignment method we've created a causal link between:



Physically active people typically have better sleep quality.

- Reduced anxiety
- Improved circadian rhythms



ARTICLE

The effect of daily walking exercise on sleep quality in healthy young adults

Wang, Feifei ; Boros, Szilvia

Sport sciences for health, 2021-06, Vol.17 (2), p.393-401; Milan: Springer Milan

“ ... Methods Fifty-four healthy adults aged 19 to 36-years old participated a pedometer based aerobic walking intervention, a cross-over randomized control trial...”



PEER REVIEWED

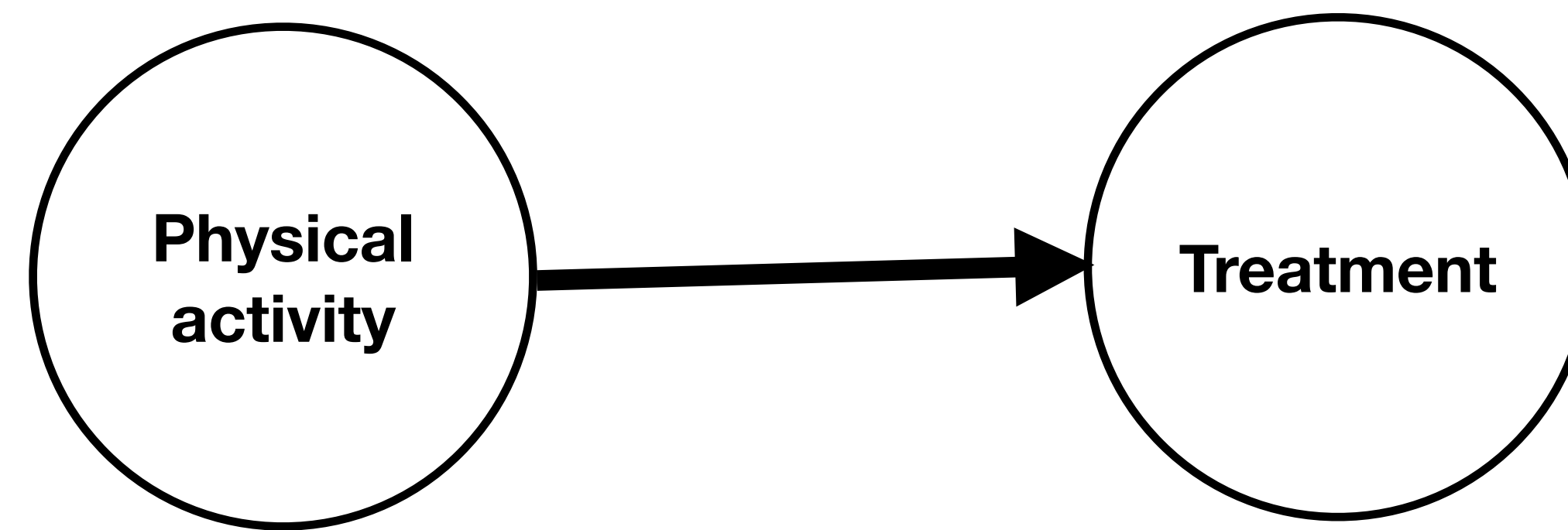


OPEN ACCESS

7.1 Randomized Experiments

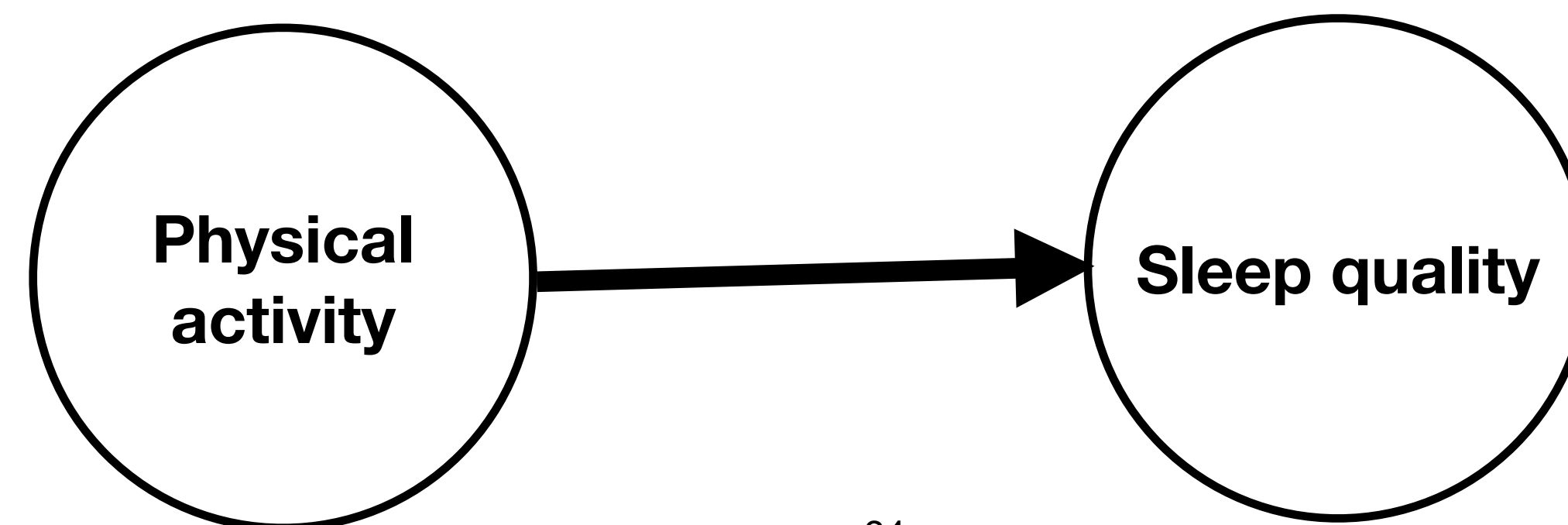
Confounders

Using this assignment method we've created a causal link between:



Physical activity influences both treatment and sleep quality.

Causal link between physical activity and sleep quality:

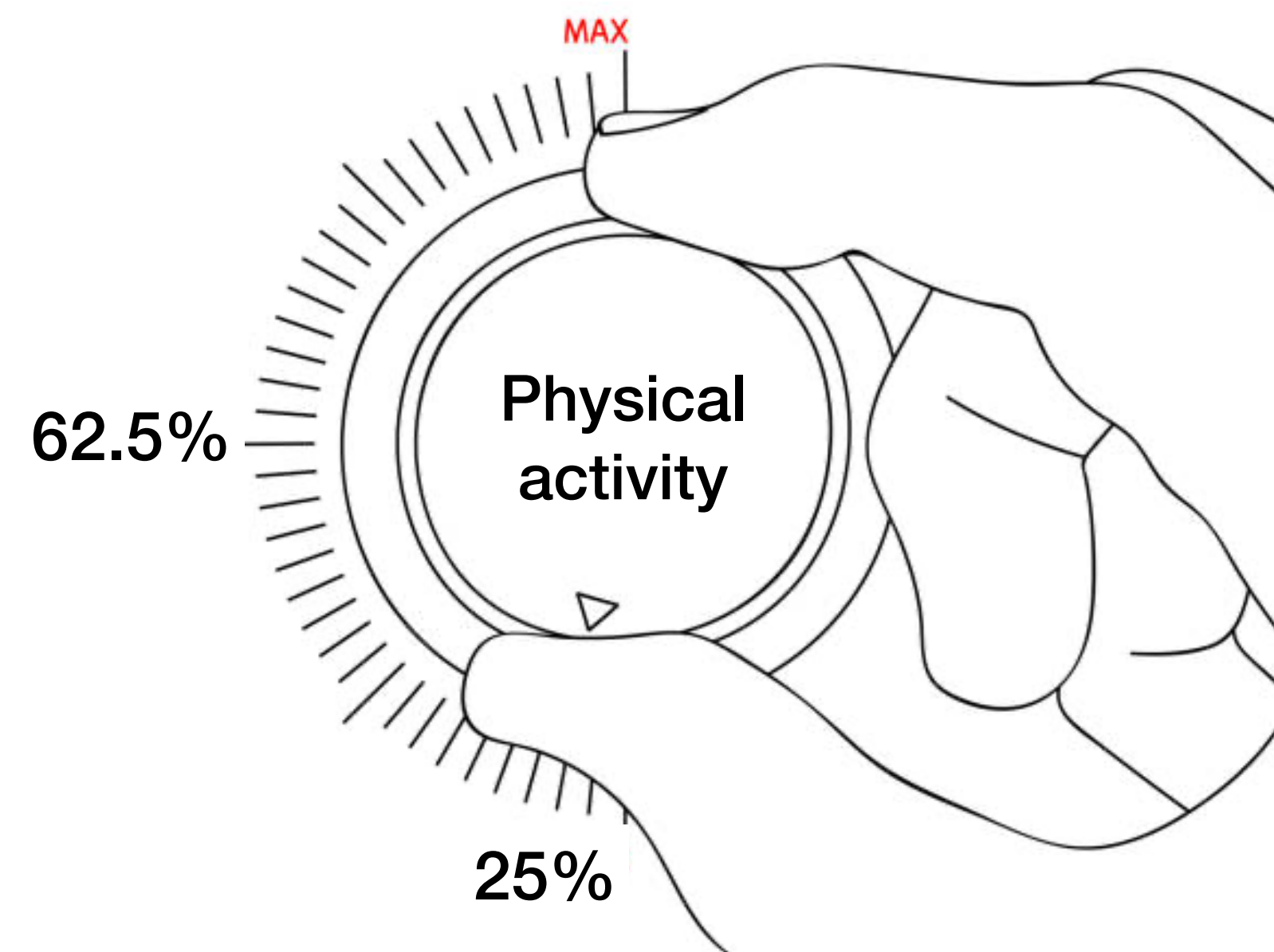


This dual influence is a **flaw in our experimental design**

7.1 Randomized Experiments

Confounders

- Because of causal link between physical activity and treatment, the physical activity dial is turned at the same time we flick the switch on.
- We don't know what caused the response we observed.



7.1 Randomized Experiments

Confounders

Our experiment is ruined.



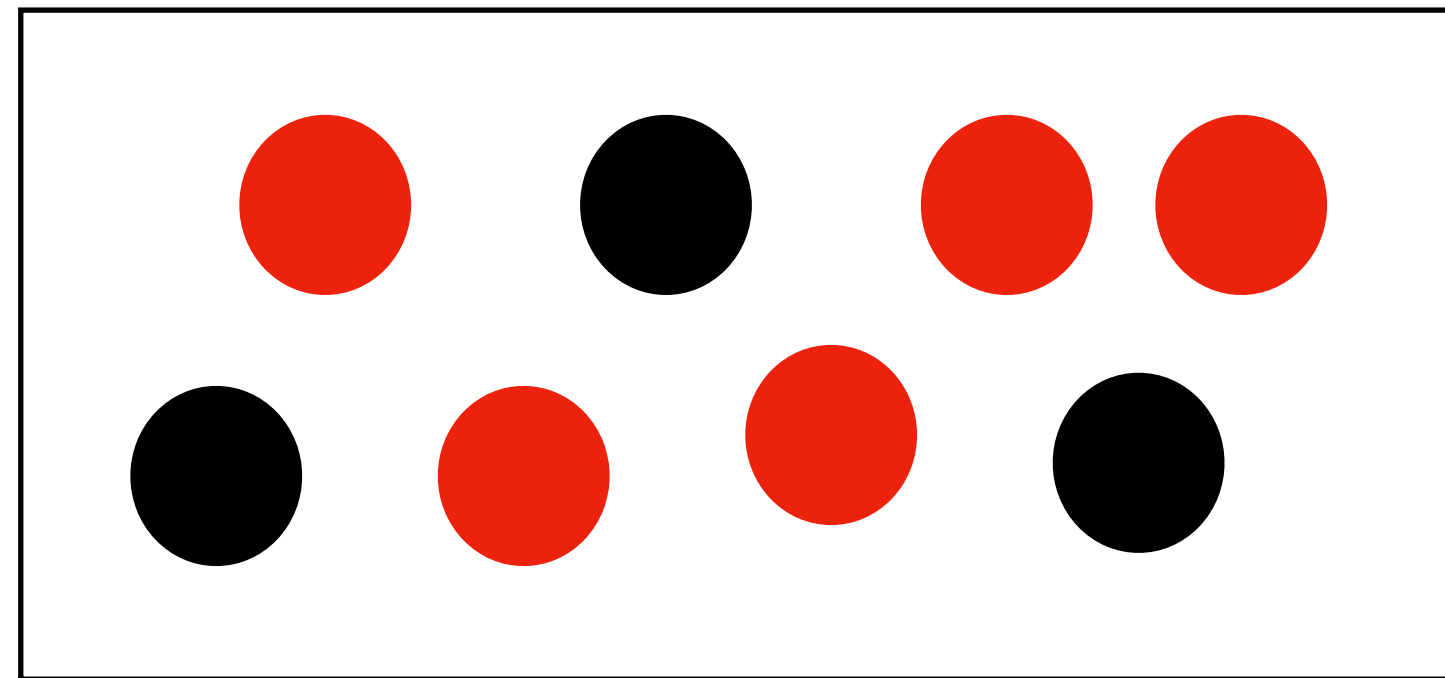
7.1 Randomized Experiments

Confounders

What's the conclusion of our experiment?

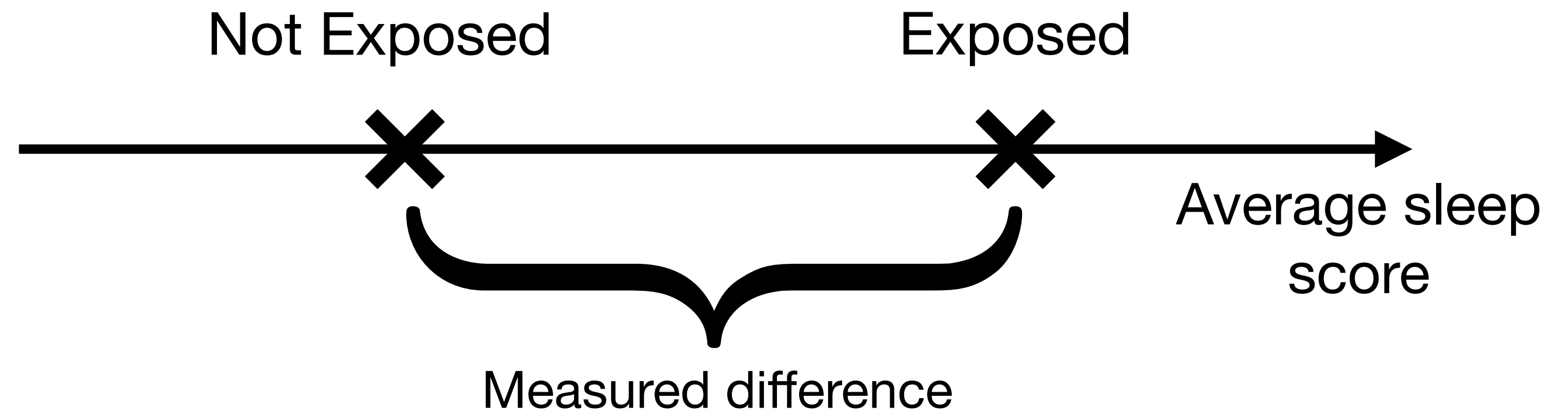
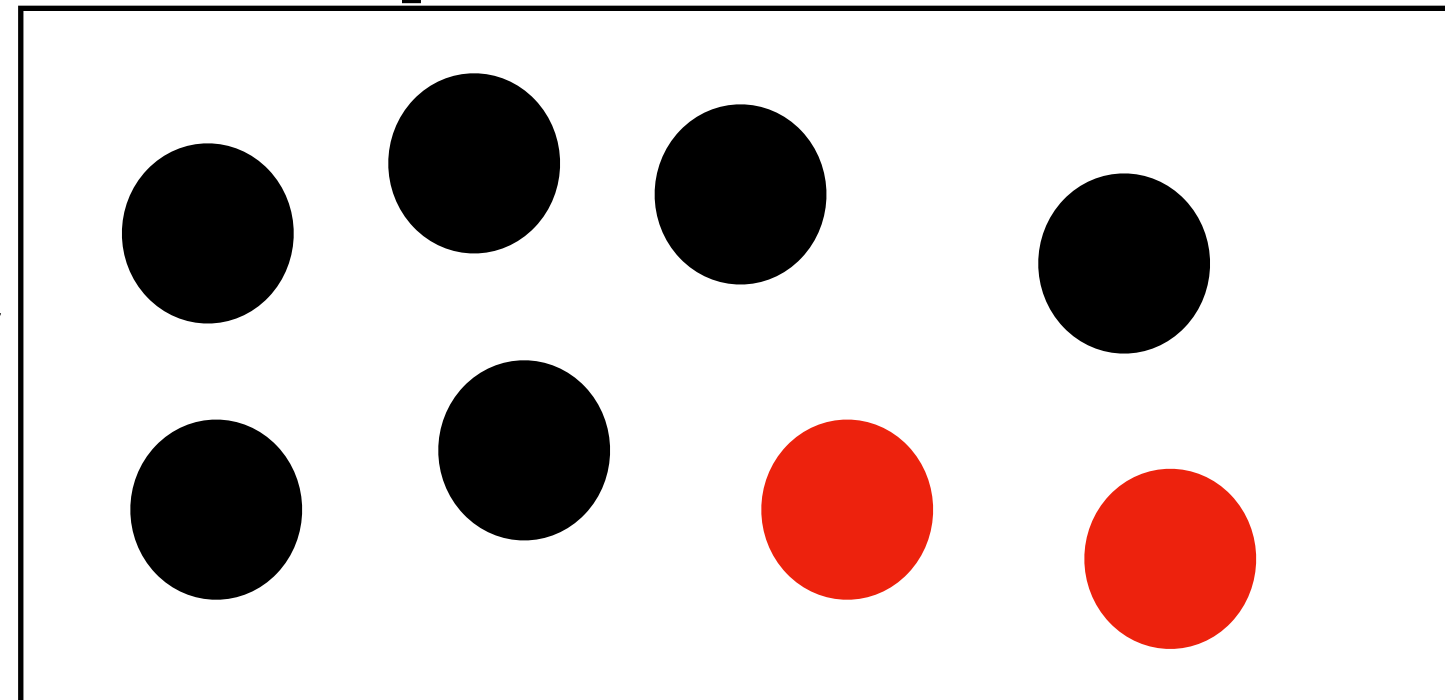
Daylight Exposure

62.5%
physically
active



No exposure

25%
physically
active



What is the increase in sleep score due to?

- Daylight exposure ?
- Physical activity ?
- Or a combination of daylight and physical activity?

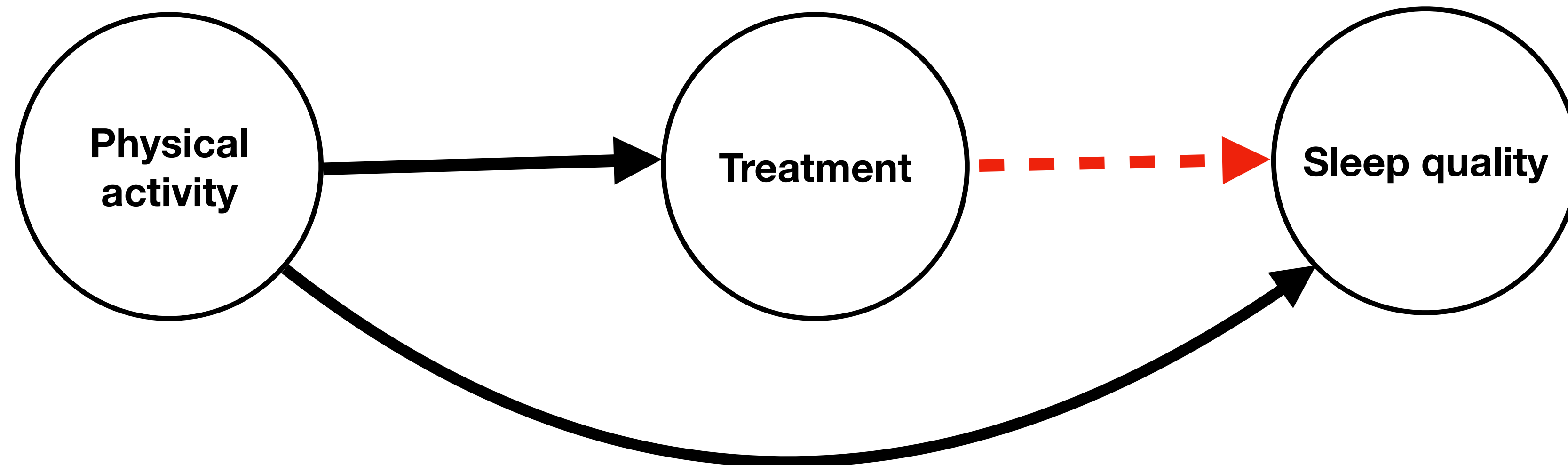
We don't know!

physical activity: Higher sleep scores

7.1 Randomized Experiments

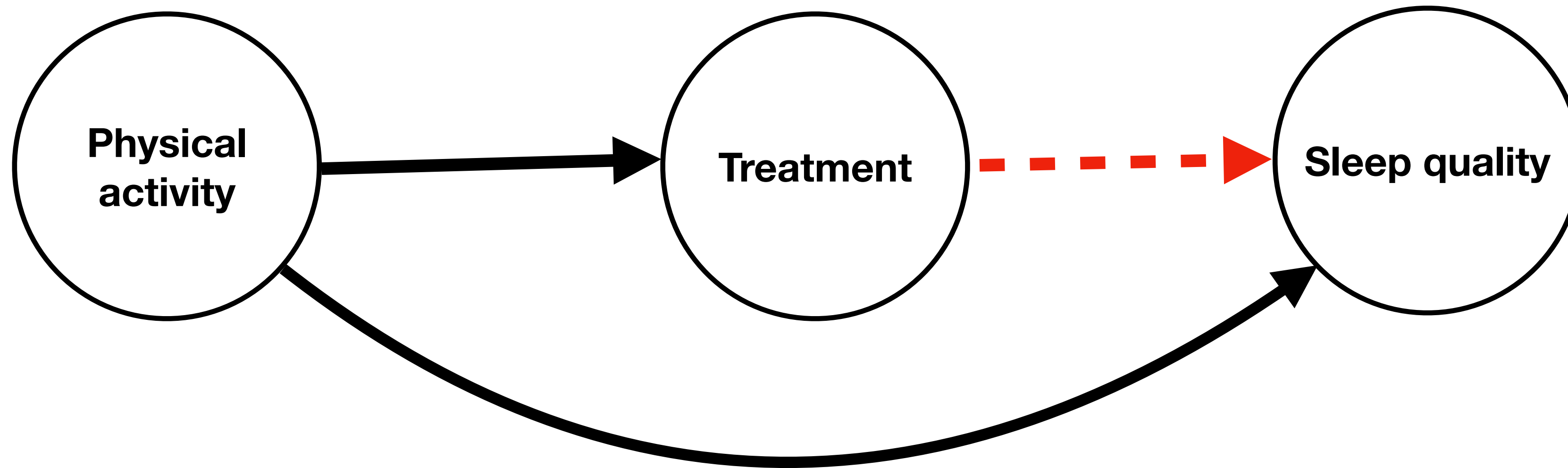
Confounders

- Physical activity is what is called a **confounder** in this experiment
- It confounds or mixes up the truth!



7.1 Randomized Experiments

Confounders



- A confounder influences **both** the dependent variable (e.g. sleep quality) and the independent variable (e.g. treatment).

It distorts our perception of the relationship between treatment and sleep quality.

- No relationship between sleep quality and sun exposure -> appear as if there is one.
- Weak relationship between sleep quality and sun exposure -> appear as if the relationship is stronger than it is

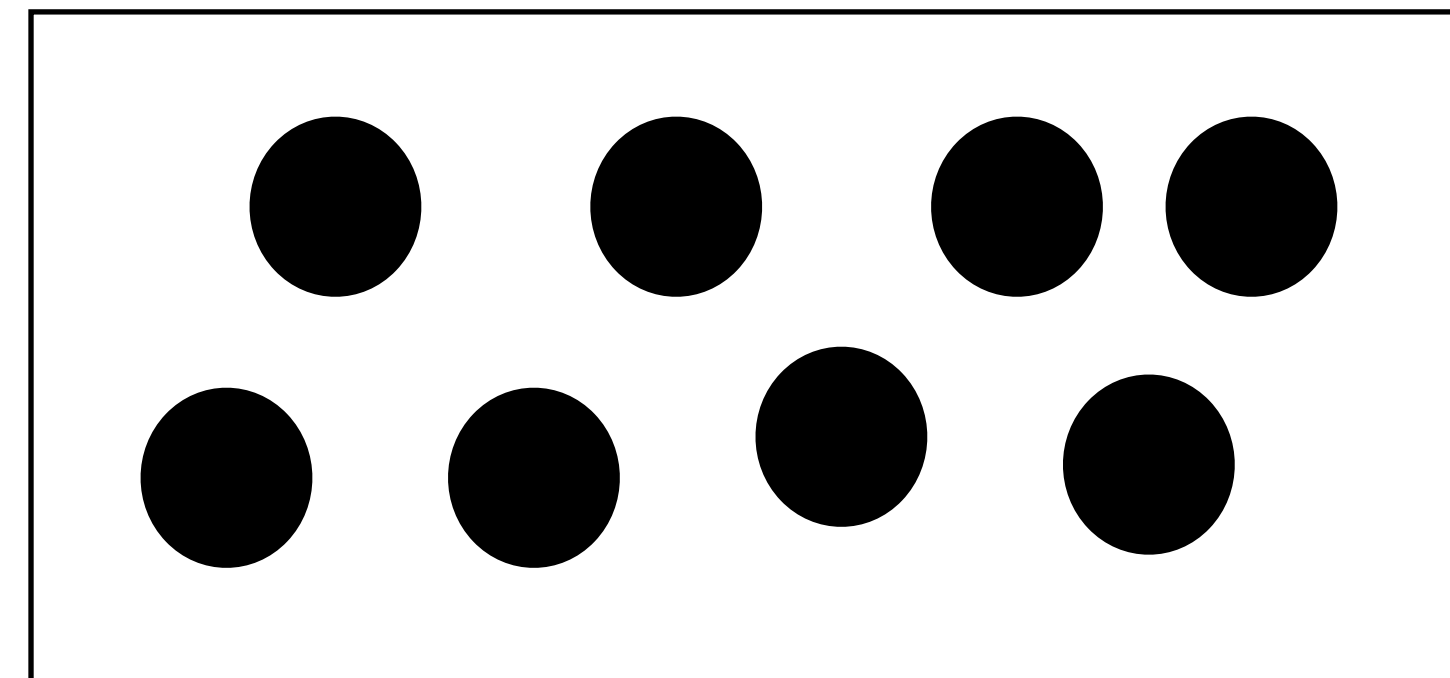
7.1 Randomized Experiments

Confounders

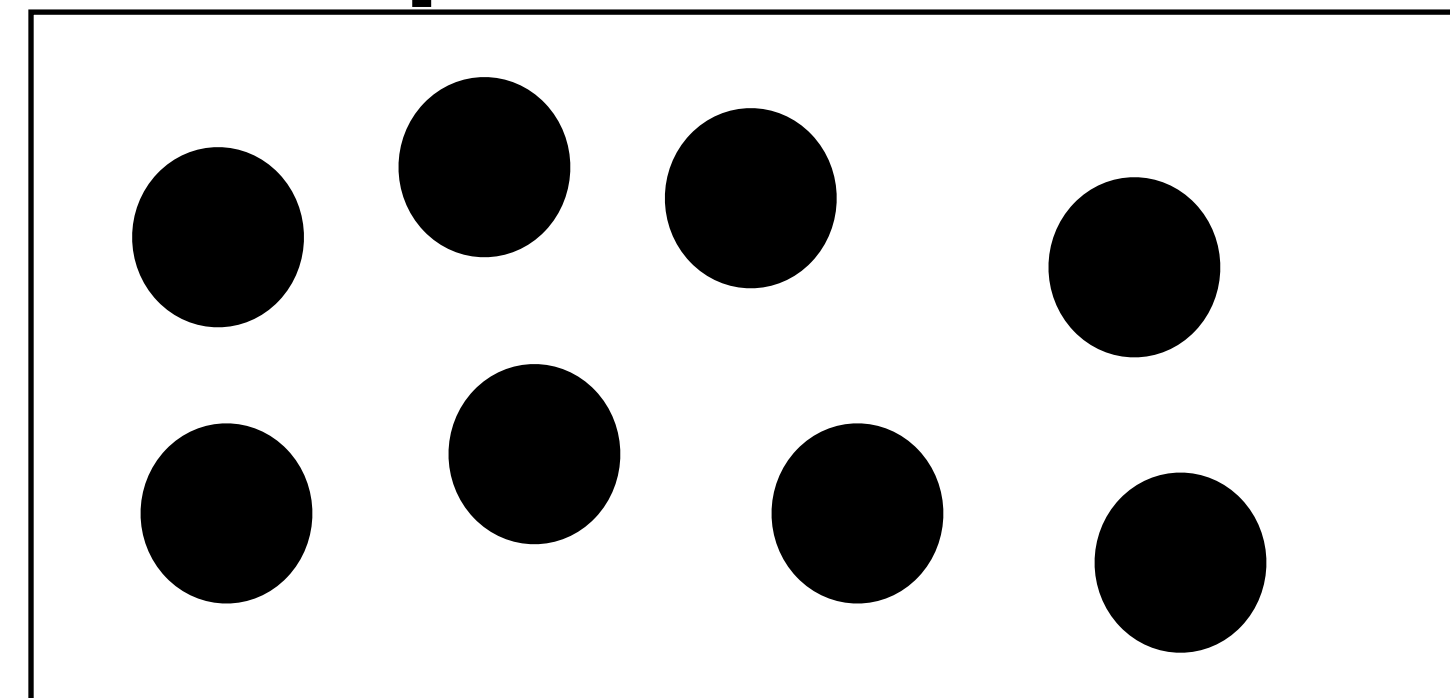
Example 2: Assign the first 8 people to pick up their Fitbit to the ‘no exposure’ group and the rest to the ‘Daylight exposure’ group.

How might the people in the **exposure** group be different from the **no exposure** group?

Daylight Exposure



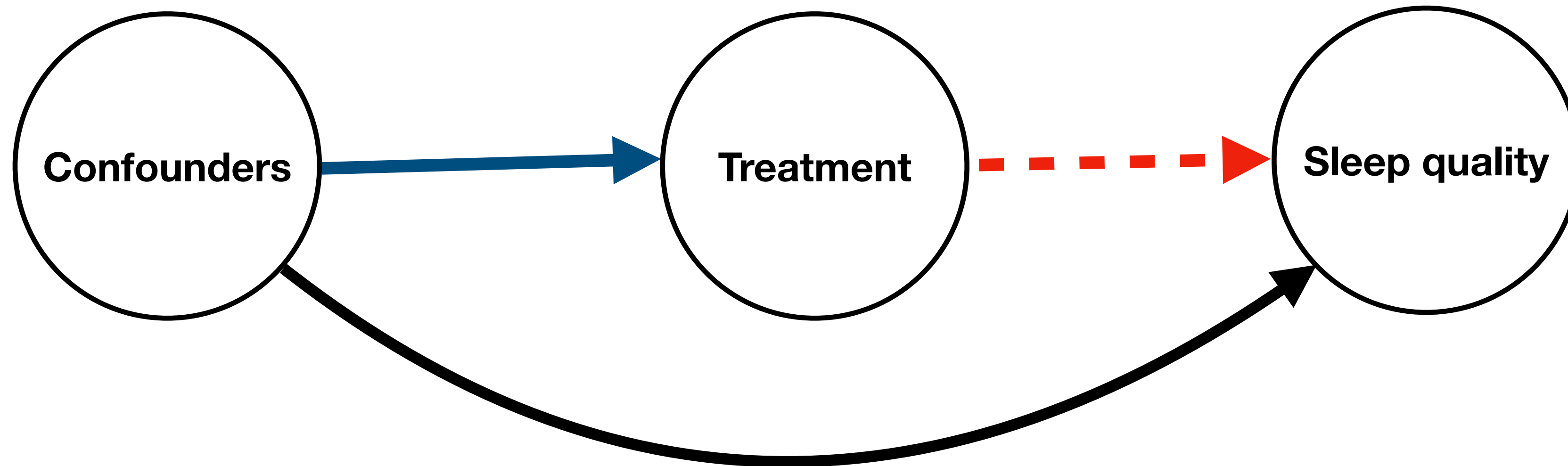
No exposure



7.1 Randomized Experiments

Confounders

- In this experiment, there are many factors that are going to affect sleep quality.
- To measure the effect of treatment on sleep quality, we need to find a way to **remove the causal link between any confounders and treatment.**



7.1 Randomized Experiments

Randomization

We need a way to assign units to treatments that protects us against both **known** and **unknown** confounders

Randomization is a solution!

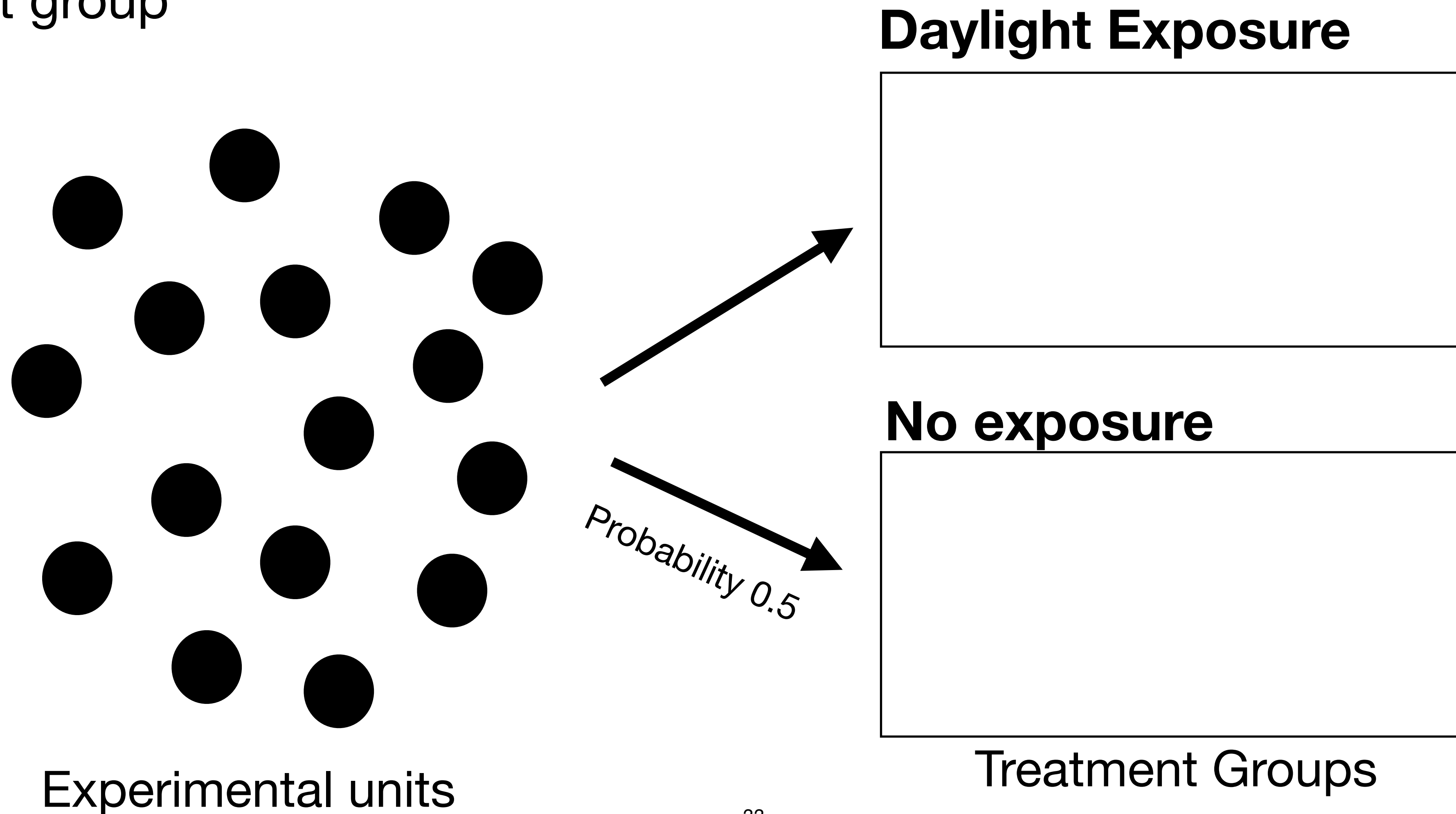
Randomization is a process of randomly assigning treatments to units.

Each unit, regardless of its features, has an equal probabilities of being assigned a treatment.

7.1 Randomized Experiments

Randomization

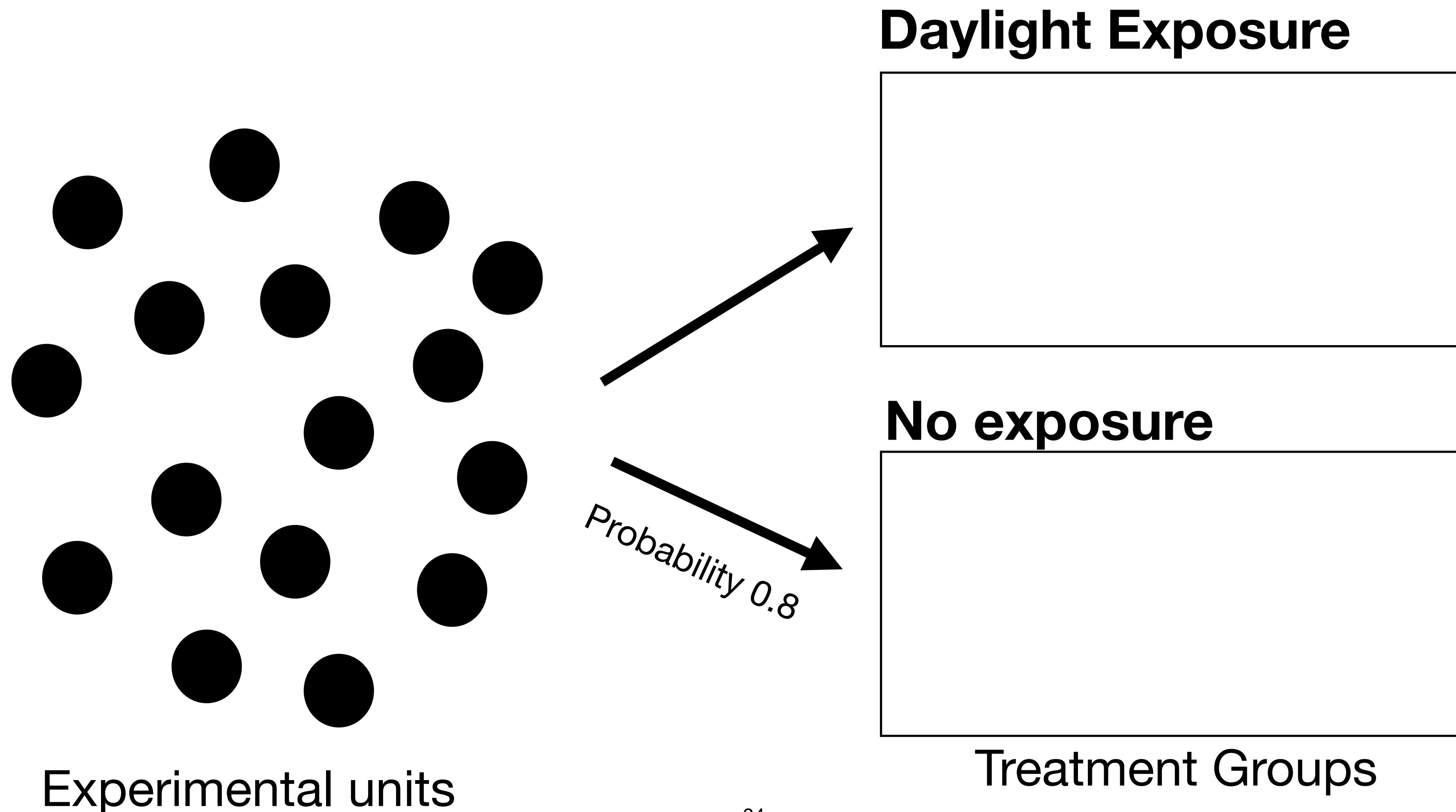
Example: Every unit has a 50% chance of being assigned to either treatment group



7.1 Randomized Experiments

Randomization

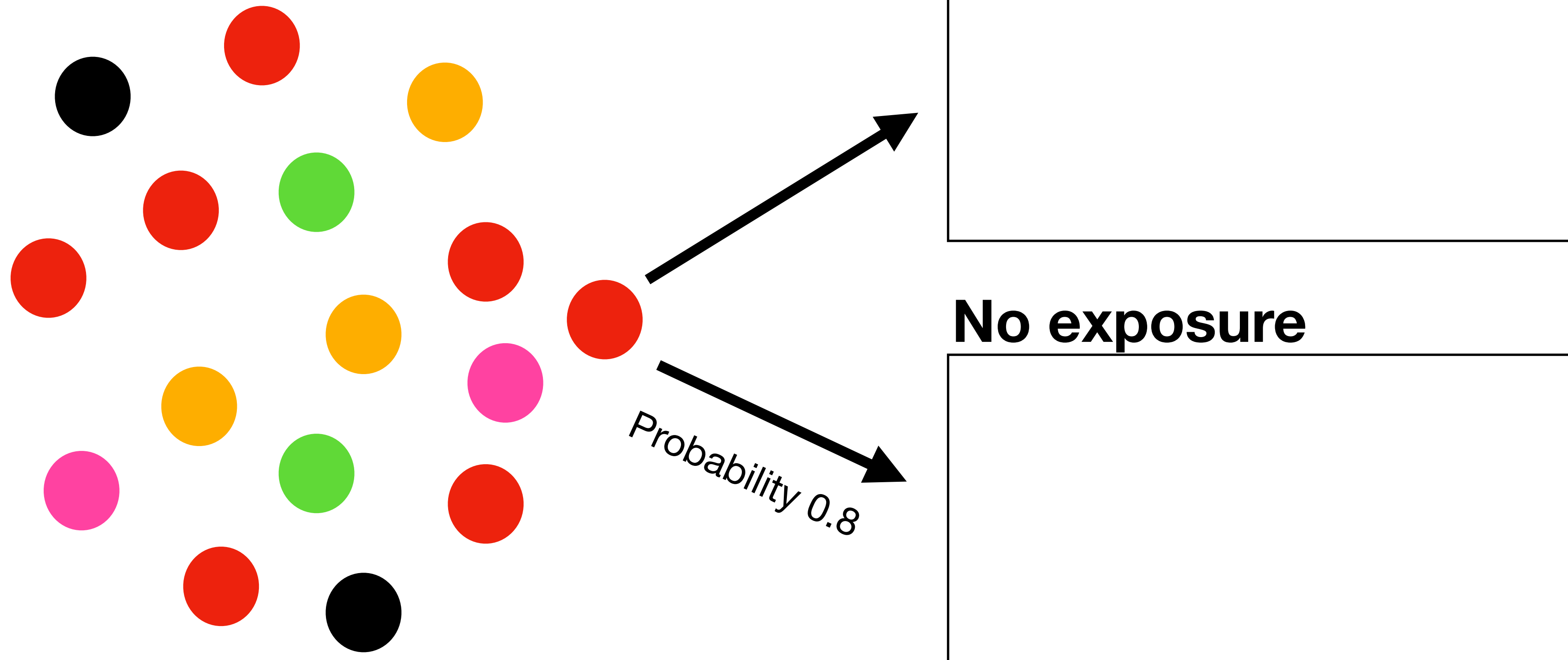
Example: Every unit has a 20% chance of being assigned to **exposure** group



7.1 Randomized Experiments

Randomization

Every unit is treated the same.
Regardless of what features they may possess.

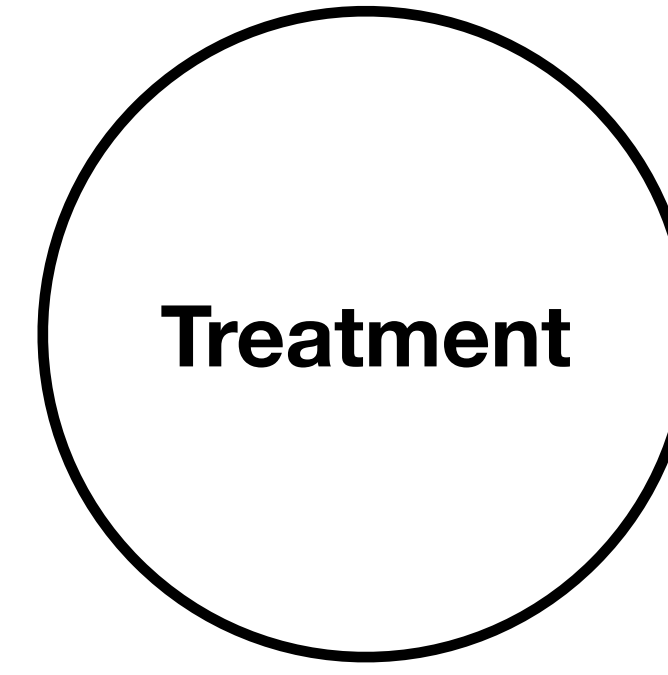
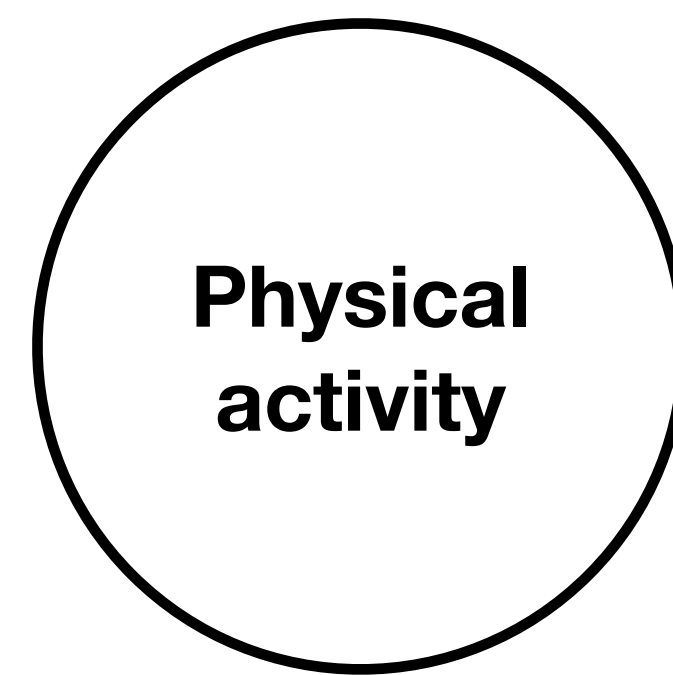


7.1 Randomized Experiments

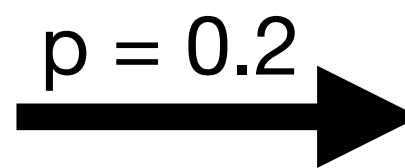
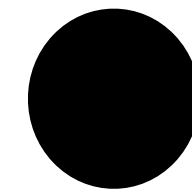
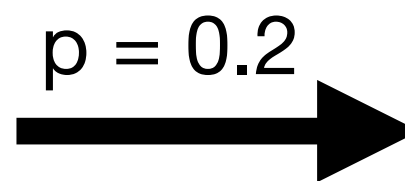
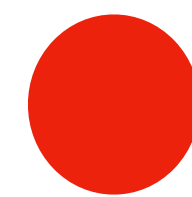
Randomization

Using randomization, there's no causal link between any feature and treatment:

Example:



Example:



Daylight Exposure



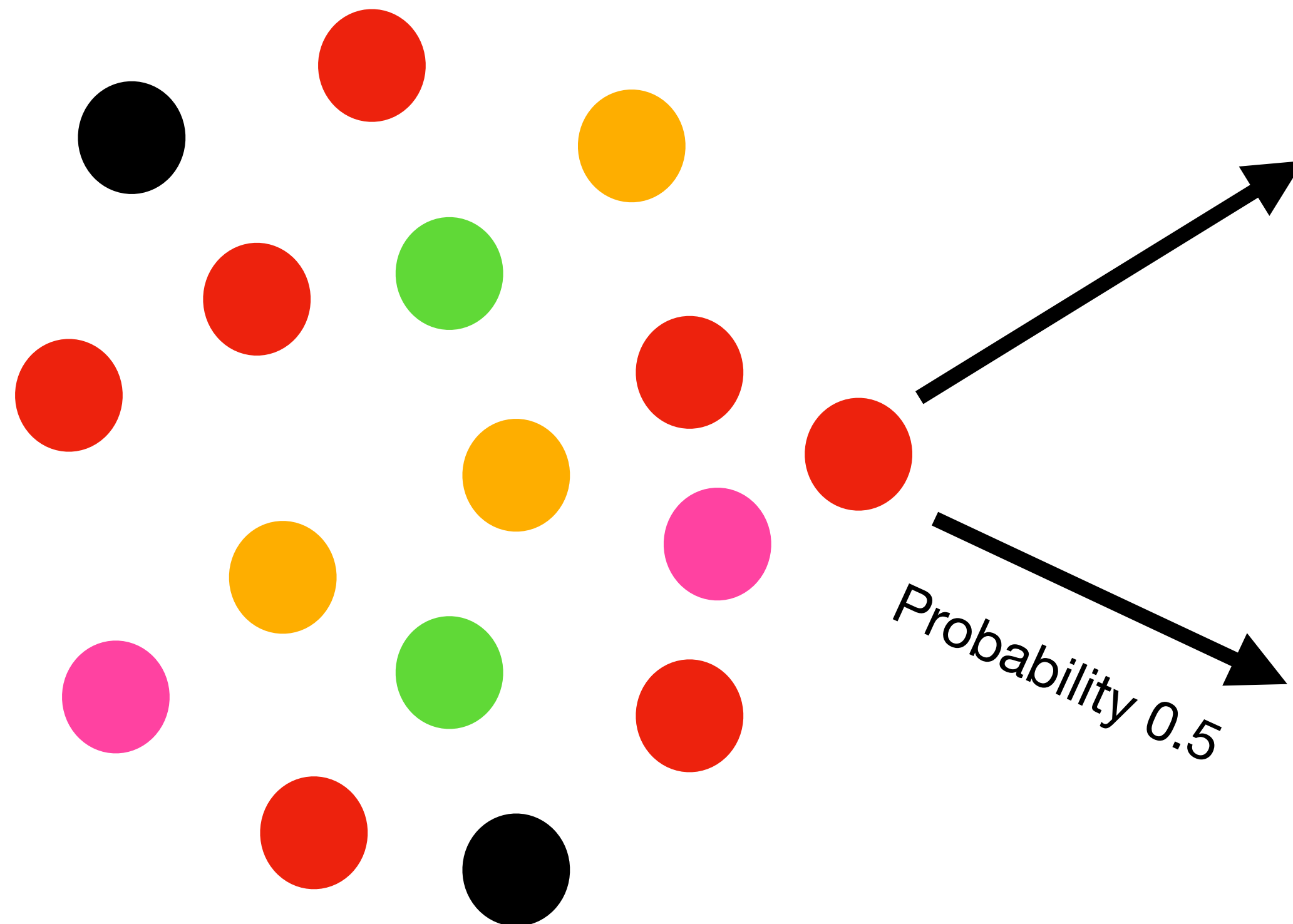
Applies to unknown or known features that may or may not be related to response!

7.1 Randomized Experiments

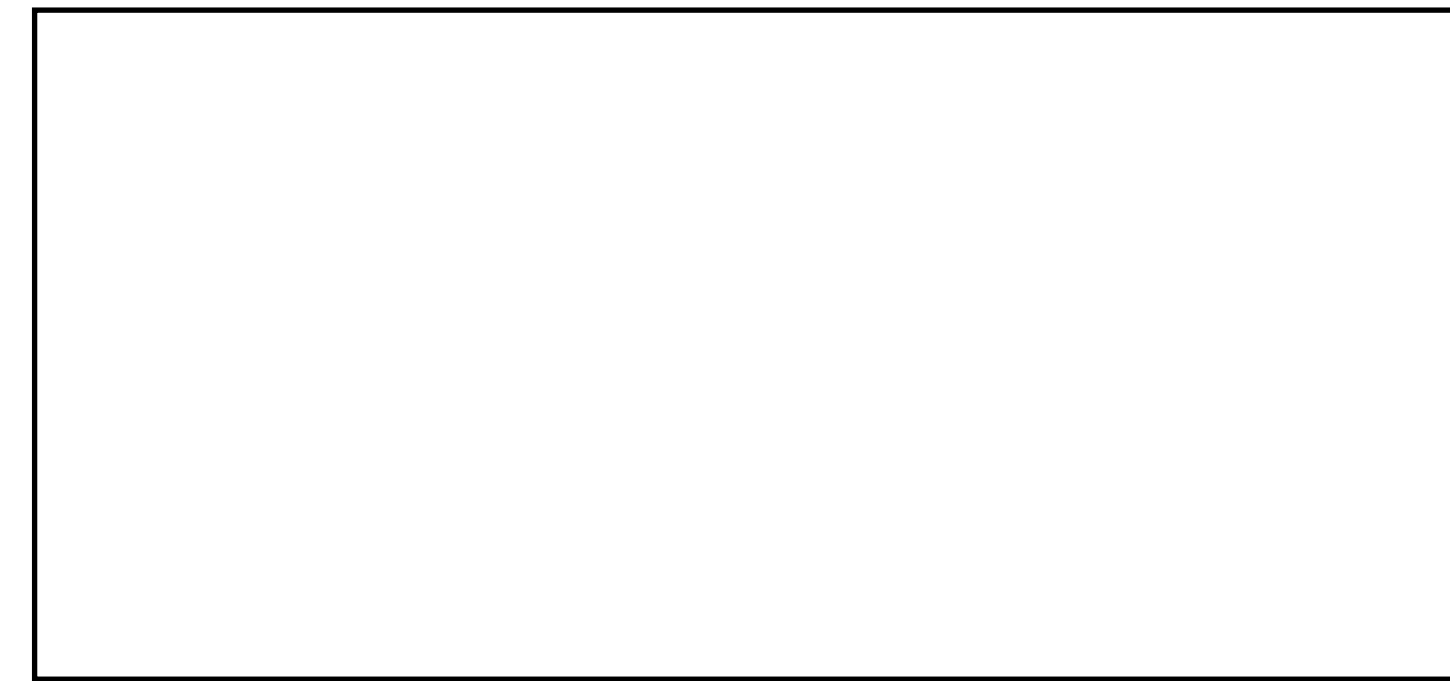
Randomization

Result of randomization:

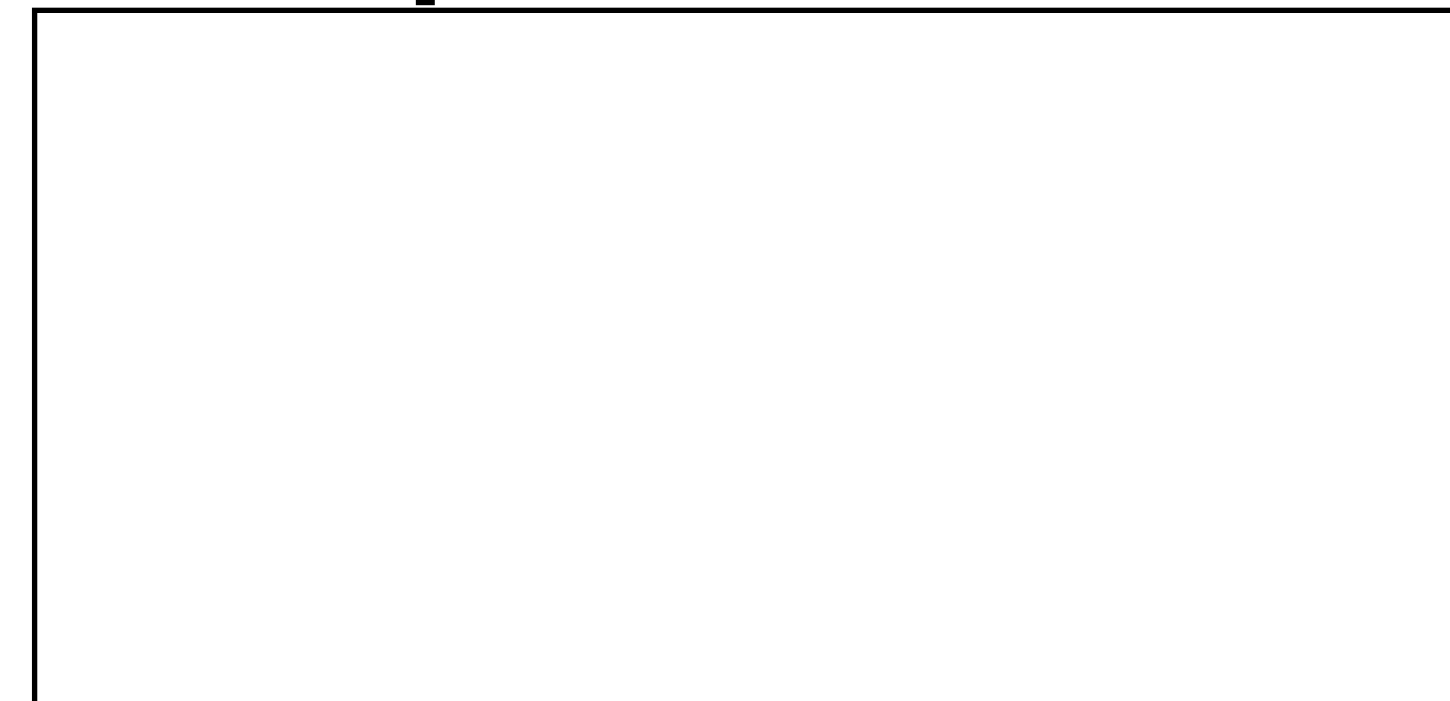
Randomize these units into groups.



Daylight Exposure



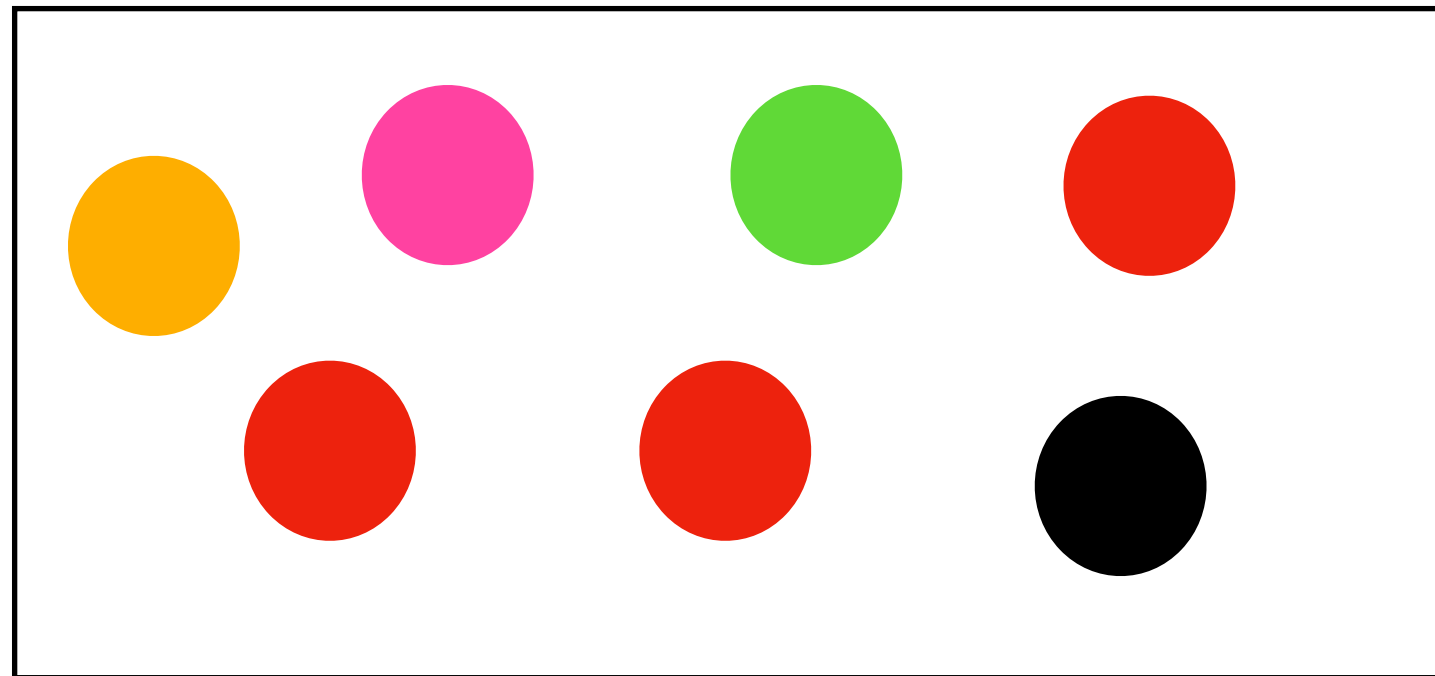
No exposure



7.1 Randomized Experiments

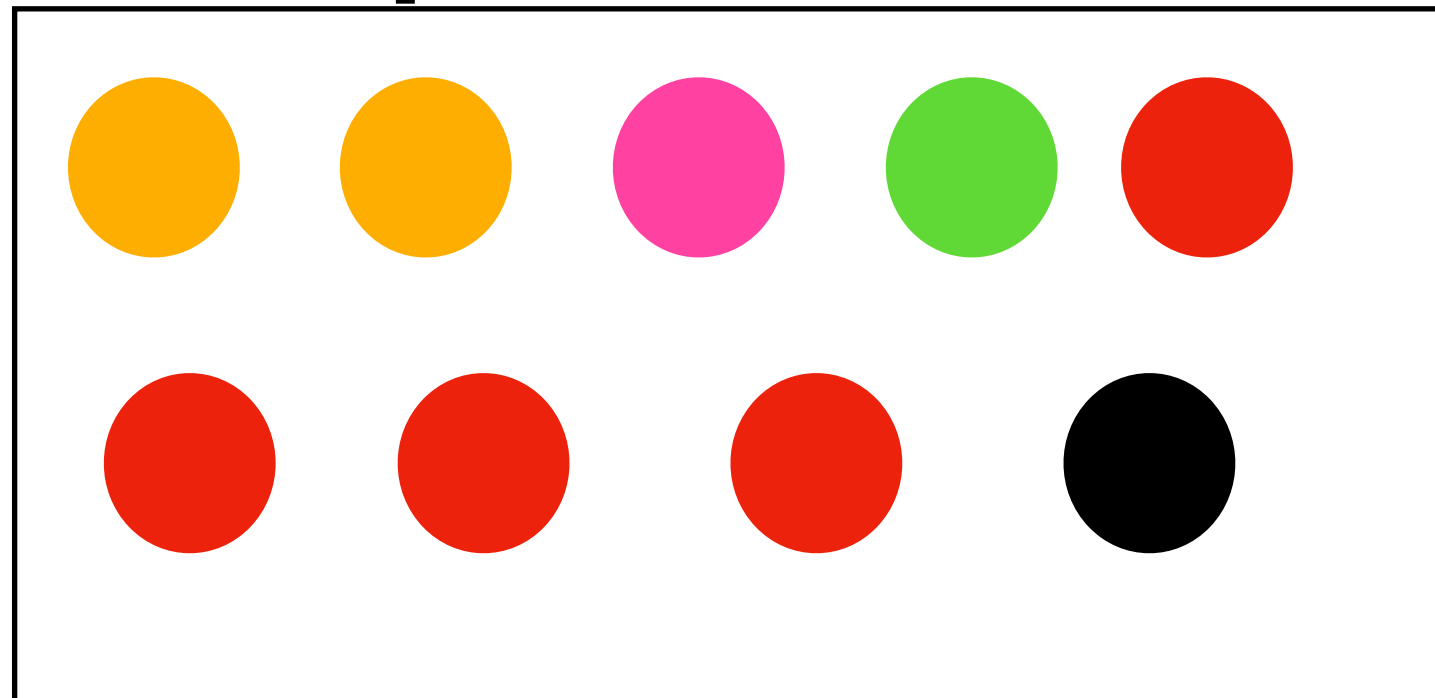
Randomization

Daylight Exposure



43% red
14% green
14% pink
14% black
14% yellow

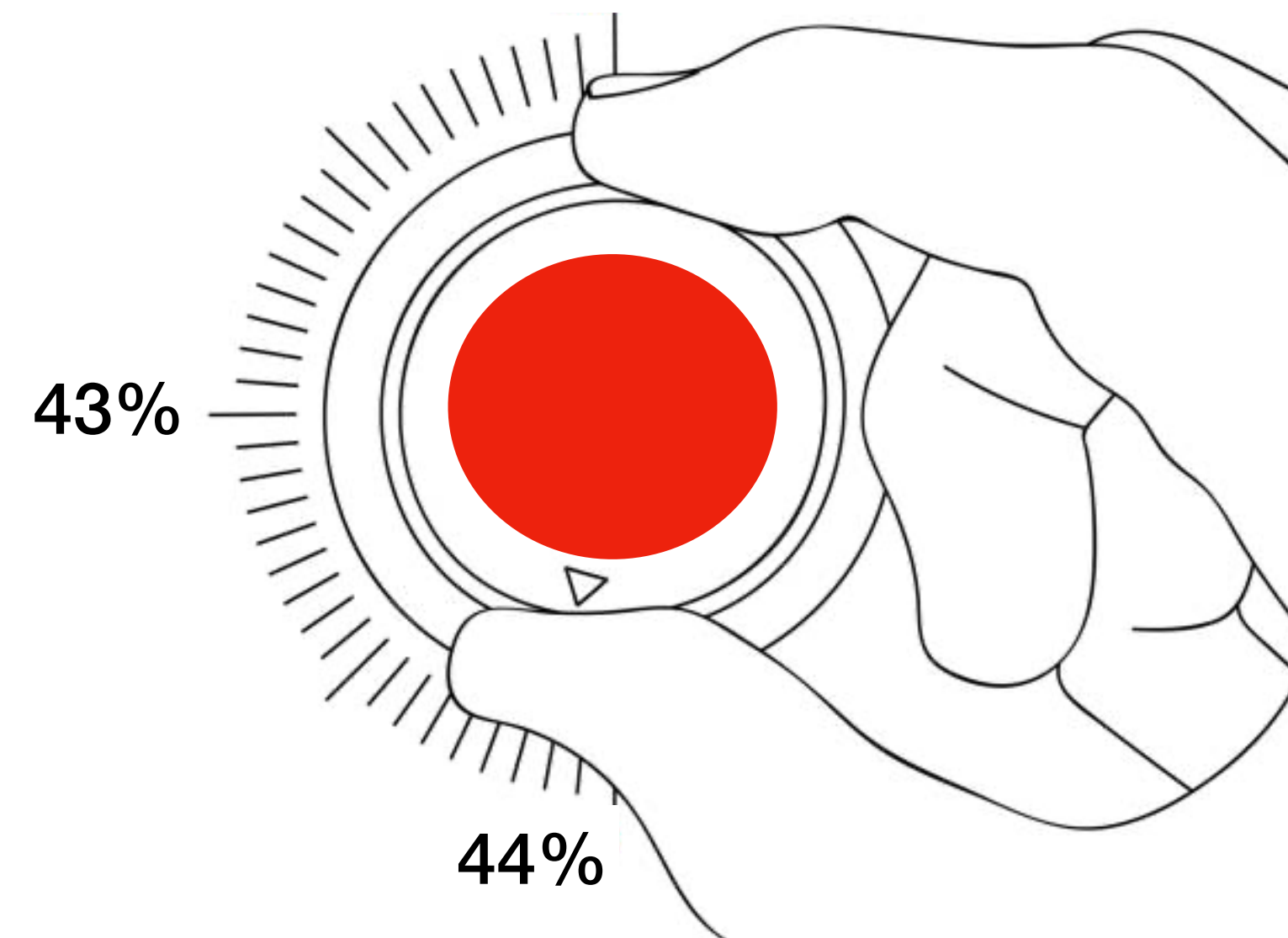
No exposure



44% red
11% green
11% pink
11% black
22% yellow

Randomized Assignment (50/50)

- Randomization helps **balance out feature proportions between groups**
- Think about the dial:

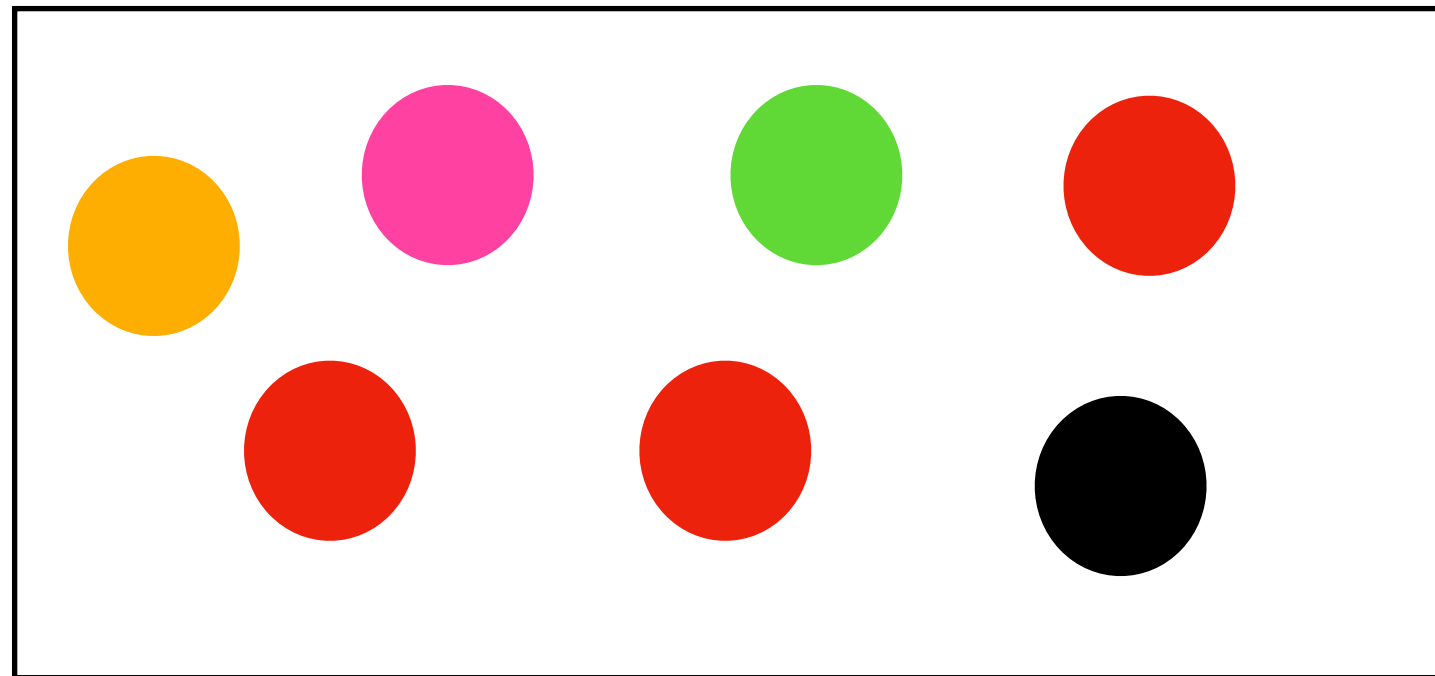


Reduce potential impact of features disproportions!

7.1 Randomized Experiments

Randomization

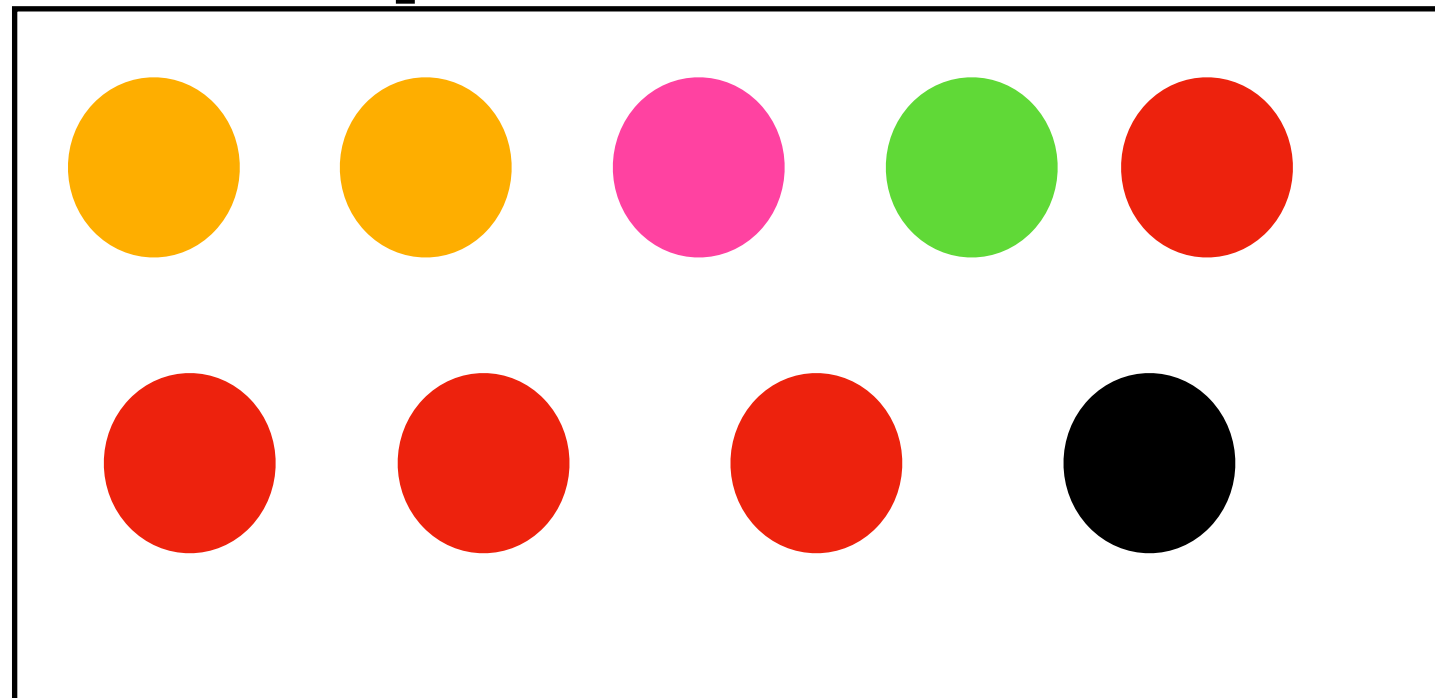
Daylight Exposure



42% red
14% green
14% pink
14% black
14% yellow

- Randomization helps **balance out feature proportions between groups**
- This balancing will improve with larger sample size.

No exposure



44% red
11% green
11% pink
11% black
22% yellow

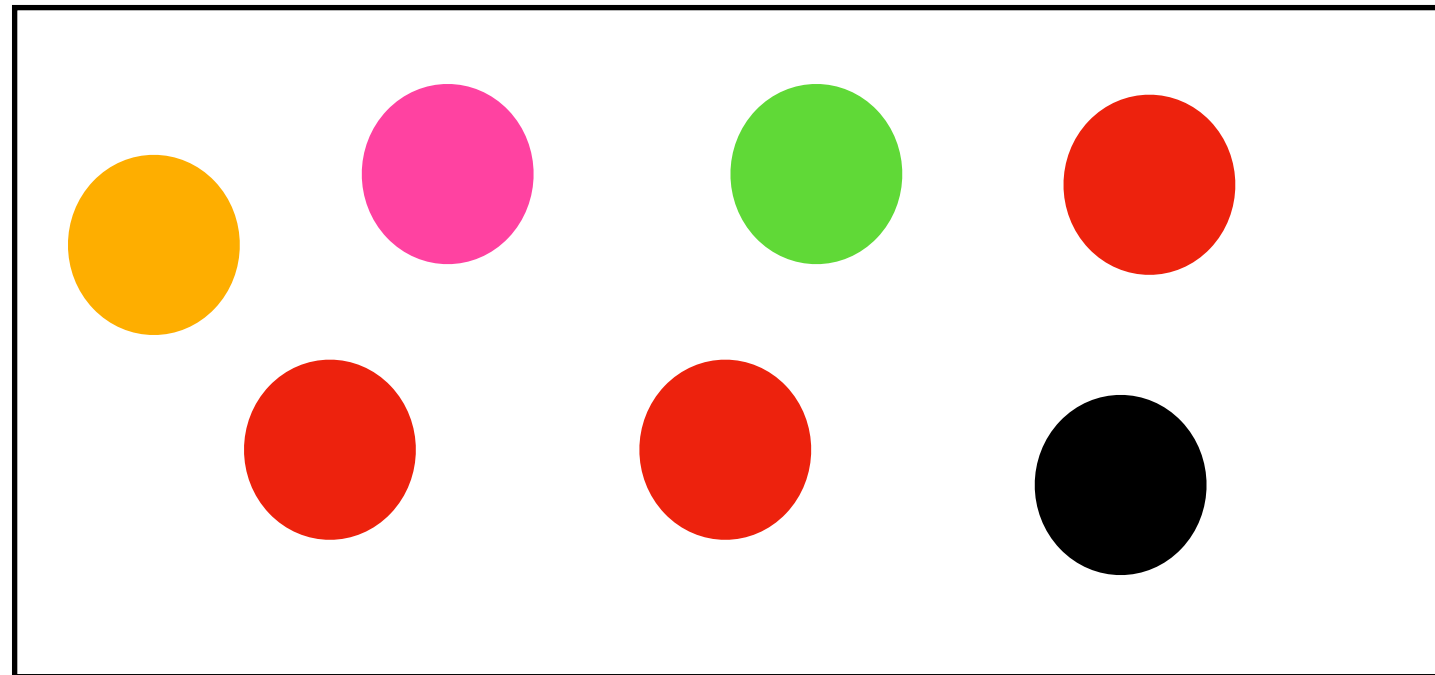
- With this type of randomization, it is not guaranteed to have treatment groups with same number of units!

Randomized Assignment (50/50)

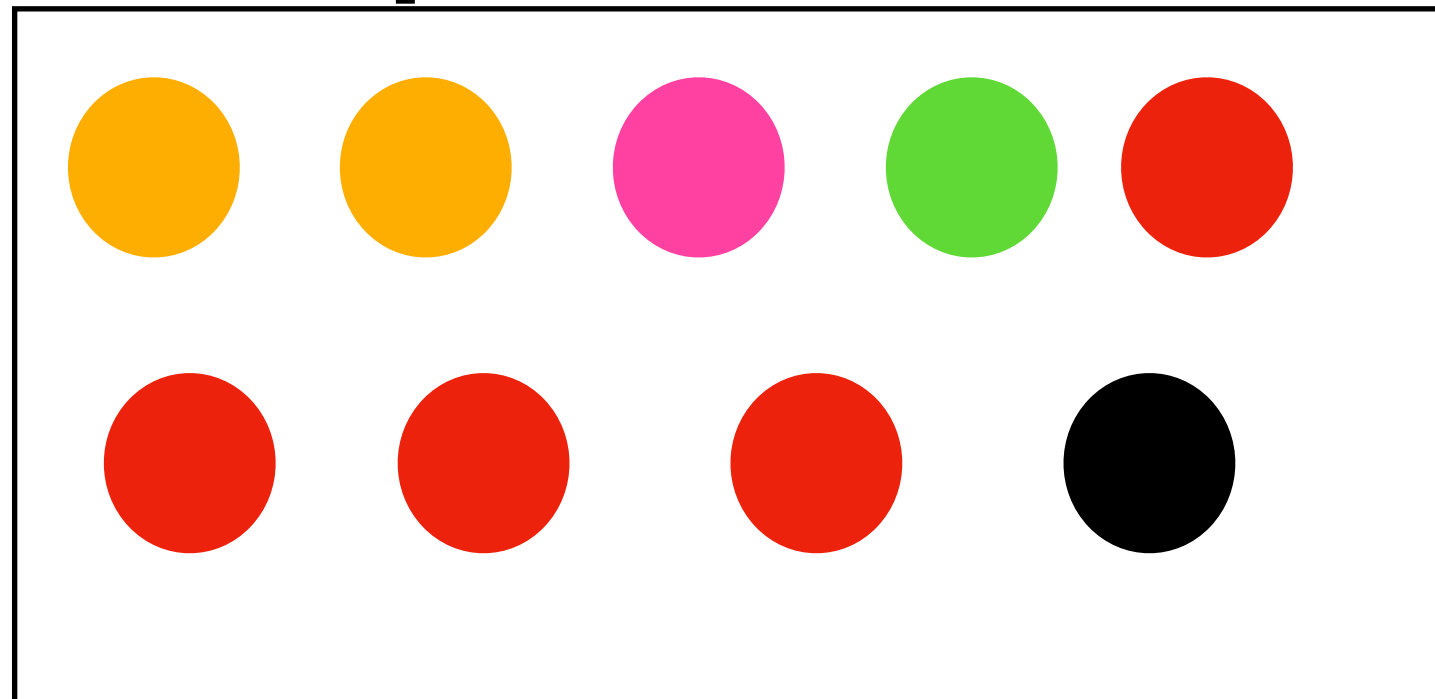
7.1 Randomized Experiments

Randomization

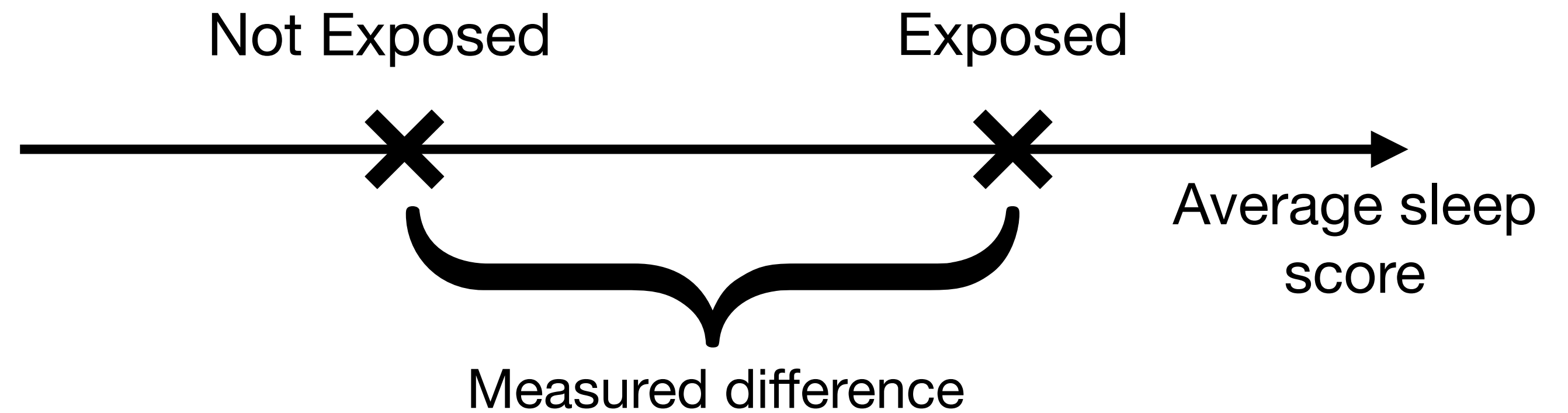
Daylight Exposure



No exposure



Randomized Assignment (50/50)



- NOW this measured difference is a better estimate of the causal effect of daylight exposure on sleep quality.

7.2 Observational data

Why we conduct surveys

- We might want to estimate a population mean, total, proportion.
- Sometimes we care about measuring relationships between two variables.
 - Hours studied and test score
 - Number of cats and happiness scores
 - Sunshine exposure and quality of sleep

7.2 Observational data

Observational studies vs experimental studies

- Observational studies
 - An observational study is an empirical study of the effects caused by treatments where the **treatment assignment mechanism is unknown**.
- This means we can only measure perceived relationships between two variables.
 - But you don't know if this relationship is due to some unobserved confounder.
 - Therefore you can only talk about **associations** between two variables instead of causation.

7.2 Observational data

Observational studies vs experimental studies

- Questions you can ask in observational data:
 - Mean, totals, and proportions of the population.
 - Associations between two variables.
 - Is the number of hours studied associated with increased test scores.
 - Is a specific diet associated with lower risk of cardiovascular disease.
- No treatment was randomly assigned. Instead we probably asked diet related questions in our survey, or number of hours studied etc..

7.2 Observational data

Concluding remarks

- Experiments are the gold standard to answer scientific questions but it's not always possible to conduct experiments.
 - Expensive
 - Unethical (assigning a unit to smoke or not smoke)
 - Takes awhile
 - Go through REB applications, recruit participants, follow-up for a few years, etc..

7.2 Observational data

Concluding remarks

- Observational data is extremely powerful as well.
- And there is ALOT of observational data out there.
- You just have to be mindful of what kind of research questions you are trying to answer and what conclusions you make.
 - Be sure to avoid using causal language.
 - Try to account for as many confounders as possible.

7.2 Observational data

Concluding remarks

- Survey studies typically produce observational data.
- You are rarely using surveys to answer causal questions.
- Survey studies:
 - Mindful of causal language.
 - You can adjust for observed confounders.
 - Mindful of non-response and other selection biases.

8.0 Nonresponse

Introduction

- We started this course by examining discussing sources of selection bias
- Sources of selection bias:
 - Undercoverage
 - Overcoverage
 - Convenience sampling
 - Self-selection
 - Non-response

8.0 Nonresponse

Introduction

- Recall that bias measures a discrepancy between the target population and the sampled population.
- Non-response:
 - **Units are sampled** but do not provide data (or provide partial data)
 - If not adjusted for, there will be bias when units that do not respond differ from people who do respond.
- For example: On an income survey those who do respond have a higher income than those who don't respond. The sample average will overestimate the population mean income.

8.0 Nonresponse

Introduction

- Two types of nonresponse:
 - 1. Unit nonresponse:**
 - The entire observation unit is missing.
 - Example: Didn't fill in the survey at all, hung up immediately, didn't return the census, etc...
 - 2. Item nonresponse:**
 - Some measurements for the observation unit is available but some are missing.
 - Example: Skipped a question in the survey or phone interview, doesn't finish a survey

8.0 Nonresponse

What to do about nonresponse

1. **Prevent it** through careful design (best way).
2. Take a representative subsample of the nonrespondents
3. Use a statistical model to adjust for nonresponse (**next class**)
 - Predict values for nonrespondents
 - Imputation models
 - Weight and calibration methods
4. Ignore nonresponse (generally not recommended)

8.1 Ignoring nonresponse

```
> airquality
  Ozone Solar.R Wind Temp Month Day
1    41    190  7.4   67     5   1
2    36    118  8.0   72     5   2
3    12    149 12.6   74     5   3
4    18    313 11.5   62     5   4
5    NA     NA 14.3   56     5   5
6    28     NA 14.9   66     5   6
7    23    299  8.6   65     5   7
8    19     99 13.8   59     5   8
9     8     19 20.1   61     5   9
10   NA    194  8.6   69     5  10
11    7     NA  6.9   74     5  11
12   16    256  9.7   69     5  12
13   11    290  9.2   66     5  13
14   14    274 10.9   68     5  14
15   18     65 13.2   58     5  15
16   14    334 11.5   64     5  16
17   34    307 12.0   66     5  17
18    6     78 18.4   57     5  18
19   30    322 11.5   68     5  19
20   11     44  9.7   62     5  20
21    1      8  9.7   59     5  21
22   11    320 16.6   73     5  22
23    4     25  9.7   61     5  23
24   32     92 12.0   61     5  24
25   NA     NA 16.6   57     5  25
```

Consider the `data(airquality)` dataset containing daily air quality measurements in New York.

The number of missing values per variable in the data is

```
colSums(is.na(airquality))
```

Ozone	Solar.R	Wind	Temp	Month	Day
37	7	0	0	0	0

8.1 Ignoring nonresponse

```
> summary(lm(Ozone~Wind, data = airquality))
```

Call:

```
lm(formula = Ozone ~ Wind, data = airquality)
```

Residuals:

Min	1Q	Median	3Q	Max
-51.572	-18.854	-4.868	15.234	90.000

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	96.8729	7.2387	13.38	< 2e-16 ***
Wind	-5.5509	0.6904	-8.04	9.27e-13 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 26.47 on 114 degrees of freedom

(37 observations deleted due to missingness)

Multiple R-squared: 0.3619, Adjusted R-squared: 0.3563

F-statistic: 64.64 on 1 and 114 DF, p-value: 9.272e-13

```
colSums(is.na(airquality))
```

Ozone	Solar.R	Wind	Temp	Month	Day
37	7	0	0	0	0

8.1 Ignoring nonresponse

```
> summary(lm(Ozone~Wind + Solar.R, data = airquality))
```

Call:

```
lm(formula = Ozone ~ Wind + Solar.R, data = airquality)
```

Residuals:

Min	1Q	Median	3Q	Max
-45.651	-18.164	-5.959	18.514	85.237

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	77.24604	9.06751	8.519	1.05e-13	***
Wind	-5.40180	0.67324	-8.024	1.34e-12	***
Solar.R	0.10035	0.02628	3.819	0.000224	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 24.92 on 108 degrees of freedom

(42 observations deleted due to missingness)

Multiple R-squared: 0.4495, Adjusted R-squared: 0.4393

F-statistic: 44.09 on 2 and 108 DF, p-value: 1.003e-14

```
colSums(is.na(airquality))
```

Ozone	Solar.R	Wind	Temp	Month	Day
37	7	0	0	0	0

8.1 Ignoring nonresponse

- Some cases have a value for ozone and a missing value for Solar.R
- Some cases have a value for Solar.R but a missing value for Ozone
- `lm(Ozone~ Wind + Solar.R, data = airquality)` will delete 42 cases from the data
- Are we making the best use of costly collected data?
- Do we have enough data now to detect the effect of interest?

```
> airquality
```

	Ozone	Solar.R	Wind	Temp	Month	Day
1	41	190	7.4	67	5	1
2	36	118	8.0	72	5	2
3	12	149	12.6	74	5	3
4	18	313	11.5	62	5	4
5	NA	NA	14.3	56	5	5
6	28	NA	14.9	66	5	6
7	23	299	8.6	65	5	7
8	19	99	13.8	59	5	8
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12	16	256	9.7	69	5	12
13	11	290	9.2	66	5	13
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16	14	334	11.5	64	5	16
17	34	307	12.0	66	5	17
18	6	78	18.4	57	5	18
19	30	322	11.5	68	5	19
20	11	44	9.7	62	5	20
21	1	8	9.7	59	5	21
22	11	320	16.6	73	5	22
23	4	25	9.7	61	5	23
24	32	92	12.0	61	5	24
25	NA	66	16.6	57	5	25

8.1 Ignoring nonresponse

```
> summary(lm(Ozone~Wind + Solar.R, data = airquality))
```

Call:

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lm(formula = Ozone ~ Wind + Solar.R, data = airquality)
```

Residuals:

Min	1Q	Median	3Q	Max
-45.651	-18.164	-5.959	18.514	85.237

Coefficients:

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Residuals:

Min	1Q	Median	3Q	Max
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Coefficients:

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(Intercept)	96.8729	7.2387	13.38	< 2e-16 ***
Wind	-5.5509	0.6904	-8.04	9.27e-13 ***

- Did the coefficient for Wind change because it was adjusted for Solar.R?
- Or should we attribute change in coefficient to changes in the subsample? (Sample size changed from 116 to 111)
- Can we compare the regression coefficients from both models?

8.1 Ignoring nonresponse

- Nonresponse increases the variance of estimators (because the sample size is smaller than anticipated).
- But the main concern is potential bias.

8.1 Ignoring nonresponse

- Consider two artificial strata of respondents and nonrespondents

Stratum	Size	Total	Mean	Variance
Respondents	N_R	t_R	$\bar{y}_{RU}$	S_R^2
Nonrespondents	N_M	t_M	$\bar{y}_{MU}$	S_M^2
Entire population	N	t	$\bar{y}_U$	S^2

$$\bar{y}_U = \frac{N_R}{N} \bar{y}_{RU} + \frac{N_M}{N} \bar{y}_{MU}$$

←

Population mean

8.1 Ignoring nonresponse

- We don't actually observe any data from the nonrespondents.
- Let's ignore the nonresponse and analyze a sample to make inference about the population mean.
- Let $\bar{y}_R$ be an approximately unbiased estimator for the population mean of respondents.
- Its bias is:
- $E[\bar{y}_R] - \bar{y}_U =$

8.1 Ignoring nonresponse

- $E[\bar{y}_R] - \bar{y}_U \approx \frac{N_M}{N}(\bar{y}_{RU} - \bar{y}_{MU})$
- The bias is small if (1) _____ or (2) _____
- But we can never be sure of (1).

8.2 Designing surveys to reduce nonresponse

- A researcher who knows the target population well will also be able to anticipate some of the reasons for nonresponse and prevent some of it.
- You can firstly rely on previous research on surveys similar to yours to find out how response rates can be improved.

8.2 Designing surveys to reduce nonresponse

- Example: This paper examined different data collection protocols.
 - Online only
 - Online at first then follow-up giving option of online or paper.
 - Choice between online or paper.
 - Choice between online or paper. But \$10 incentive to choose online.
- On top of this, there were two levels of incentives. Low and High.
- 4 X 2 design choices.

8.2 Designing surveys to reduce nonresponse

TABLE 8.2
Results for experiment in [Example 8.2](#), from Biemer et al. (2018).

Survey Protocol	Incentive	Response Rate (%)	% Internet Respondents
Online only	\$10	32.0	100
Online only	\$20	38.9	100
Online/Choice	\$10	38.6	64
Online/Choice	\$20	42.3	71
Choice	\$10	38.8	28
Choice	\$20	43.4	27
Choice Plus	\$10	42.8	64
Choice Plus	\$20	45.0	64

Results show that the choice plus with a \$20 incentive gathered the highest response rate

8.2 Designing surveys to reduce nonresponse

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Results for experiment in [Example 8.2](#), from Biemer et al. (2018).

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Online/Choice	\$10	38.6	64
Online/Choice	\$20	42.3	71
Choice	\$10	38.8	28
Choice	\$20	43.4	27
Choice Plus	\$10	42.8	64
Choice Plus	\$20	45.0	64

However, was it bringing more data in from an already well represented group or was it bringing in more data from the under represented group?

8.2 Designing surveys to reduce nonresponse

TABLE 8.2

Results for experiment in [Example 8.2](#), from Biemer et al. (2018).

Survey Protocol	Incentive	Response Rate (%)	% Internet Respondents
Online only	\$10	32.0	100
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Online/Choice	\$10	38.6	64
Online/Choice	\$20	42.3	71
Choice	\$10	38.8	28
Choice	\$20	43.4	27
Choice Plus	\$10	42.8	64
Choice Plus	\$20	45.0	64

- The study also collected demographic variables: age, race, sex, etc..
- They compared their respondents demographic to those of a census.
- Still the ‘choice plus’ protocol with the \$20 incentive had highest response rate, a high percentage of online respondents, and demographics the most similar to the census.

8.2 Designing surveys to reduce nonresponse

- So you can conduct experiments for sampling protocols or do research for studies that have already implemented similar surveys.
- There are other factors that can influence nonresponse!

8.2 Designing surveys to reduce nonresponse

- **Survey content:** The topic of the survey is more sensitive and some people may not want to answer.
 - Some solutions: Careful ordering of questions, using empathetic language in the questions, carefully word that the responses of this survey will be completely anonymous, or do something called randomized response (next class)
- **Questionnaire design:** Order and wording will affect how a person will perceive the question and will also affect whether they answer a particular question on a survey.
 - Can you think of ways questionnaire design would affect response rates?

8.2 Designing surveys to reduce nonresponse

- **Survey introduction:**
 - Telling someone in advance that they have been selected for a survey.
 - Telling them the motivation for the survey. The potential impact. Why should they care to provide you data?
 - Outlining the rights of the participants (rights to privacy, to withdraw from a study at any time)
 - Who is conducting the survey, how will their data be protected, how will their identity be preserved etc..

8.2 Designing surveys to reduce nonresponse

- **Sampling frames**
 - Some are better than others.
 - If a sampling frame has contact information which makes it easier to follow-up to individuals or businesses linked to the sampled units that aren't responding.
- **Survey mode**
 - As we have seen, depending on your target population, the mode of the survey (online, paper, mail, in-person) may have an impact on the response rate.
 - Also online, they may be able to skip questions, but in-person it's less easy.
 - Important to know who your study population is when designing a survey.

8.2 Designing surveys to reduce nonresponse

We just just described a few design choices that may help reduce nonresponse rates.

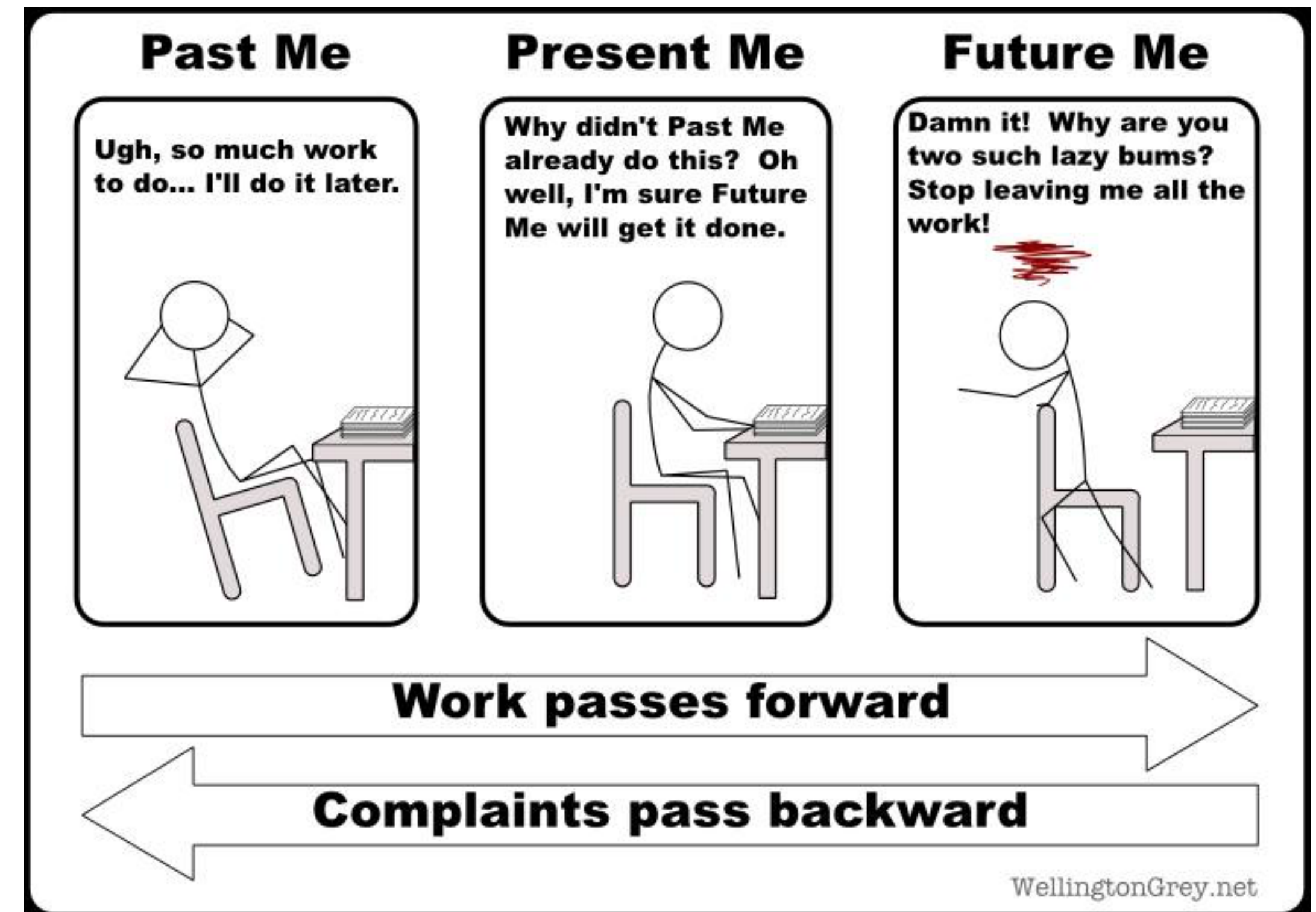
This is by no means a comprehensive list!

More ways include:

- Interviewer choices
- Burden on respondent (too many questions, questions too long to read). Remember the respondents are doing YOU a favour!
- Respondent motivations (either monetary incentives or otherwise)
- Respondent availability (phone/visit at better times of day/week, you can ask “can I call/come back at a better time?”)

8.2 Designing surveys to reduce nonresponse

- Just remember, that the quality of a survey is largely determined at the design stage.
- Having data from a nicely designed study will make any subsequent analysis a lot more straightforward and reliable.
- Any time you can spend at the design stage will be worth it and save a lot of trouble later.



8.3 Subsample of the nonrespondents

Two-Phase Sampling

- Okay okay. We did our best. We designed the best possible survey EVER!!!!
- We still have people who didn't respond.



8.3 Subsample of the nonrespondents

Two-Phase Sampling

- Round TWO of sampling the nonrespondents
- This is called the two-phase sampling approach.
- Suppose there are n_R respondents and n_M non-respondents (M = “Missing”)
- $n = n_R + n_M$
- Now let's sample a percentage ν % of n_M

8.3 Subsample of the nonrespondents

Two-Phase Sampling

- Let's sample a percentage $\nu\%$ of n_M
- Suppose that all νn_M units respond.
- Let $\bar{y}_M$ be the sample mean of those νn_M units.
- Let $\bar{y}_R$ be the sampled mean of the n_R units.
- An estimator for the population mean is:

$$\hat{\bar{y}} = \frac{n_R}{n} \bar{y}_R + \frac{n_M}{n} \bar{y}_M$$

Makes sense! We know $\bar{y}_M$ because of the second-stage sampling we did.

We assume $\bar{y}_M$ represents all the n_R originally sampled units.

8.3 Subsample of the nonrespondents

Two-Phase Sampling

- Similarly: for the total we have

$$\hat{t} = N\hat{\bar{y}} = \frac{N}{n} \sum_{i \in S_R} y_i + \frac{N}{n} \frac{1}{\nu} \sum_{i \in S_M} y_i$$

- Every unit in the sample among respondents represents itself and _____ other upsampled people in the population
- Every unit in the second-stage sample among nonrespondents represents itself and _____ sampled non-response units.
- Every unit in the sample among nonrespondents represents itself and _____ other upsampled people in the population

8.3 Subsample of the nonrespondents

Two-Phase Sampling

- The subsampling can be high for a survey where low levels of response are predicted.
- You can change the mode for the second phase.
 - Initial phase might have been a online or mail in survey.
 - In the second phase, you can sample non-respondents and visit them or send a more personalized email, etc..

8.3 Subsample of the nonrespondents

Two-Phase Sampling

- Similarly: for the total we have

$$\hat{t} = N\hat{y} = \frac{N}{n} \sum_{i \in S_R} y_i + \frac{N}{n} \frac{1}{\nu} \sum_{i \in S_M} y_i$$

Exercise: Can you show this?

8.4 Response propensities

Auxiliary information for treating nonresponse

- Suppose y_i is the characteristic of interest for unit i (measured only for respondents).
- Let $\mathbf{x}_i$ be a vector of auxiliary information for unit i
- This auxiliary information can come in different forms.

8.4 Response propensities

Auxiliary information for treating nonresponse

- This auxiliary information can come in different forms.
 - A. Known for every unit in the sampling frame. For example, a list of registered voters, used as sampling frame, may contain voter's age, sex, history of participating in previous elections, etc...

8.4 Response propensities

Auxiliary information for treating nonresponse

- This auxiliary information can come in different forms.
 - A. Known for every unit in the sampling frame. For example, a list of registered voters, used as sampling frame, may contain voter's age, sex, history of participating in previous elections, etc...
 - B. Known for the sampled units. For example, you might be interested in income but collected other information on the units like age, job title, highest level of education, etc..

8.4 Response propensities

Auxiliary information for treating nonresponse

- This auxiliary information can come in different forms.
 - A. Known for every unit in the sampling frame. For example, a list of registered voters, used as sampling frame, may contain voter's age, sex, history of participating in previous elections, etc...
 - B. Known for the sampled units. For example, you might be interested in income but collected other information on the units like age, job title, highest level of education, etc..
 - C. Population means or totals from an external data source. For example, demographics from census surveys or population registry.

8.4 Response propensities

Response Propensities

- In chapters 2 to 7 we have used the random variable I_i to indicate selection for the sample.
- $P(I_i = 1) = \pi_i$
- Now define a response random variable:

$$R_i = \begin{cases} 1 & \text{if unit } i \text{ responds} \\ 0 & \text{if unit } i \text{ does not respond} \end{cases}$$

8.4 Response propensities

Response Propensities

- After data collection, R_i will take on a value of 1 or 0.
 - If it takes on a value of 1, a value of y_i is recorded.
 - If it takes on a value of 0, a value of y_i is missing.
- Define the response propensity (propensity score) for unit i as:

$$\phi_i = P(R_i = 1)$$

8.4 Response propensities

Response Propensities

- If I_i and R_i are independent then:

$$P(\text{unit } i \text{ selected and responds}) = E[I_i R_i] = \pi_i \phi_i$$

- If we knew true propensity score ϕ_i , we could estimate the total as:

$$\hat{t}_\phi = \sum_{i=1}^N I_i R_i \frac{y_i}{\pi_i \phi_i} = \sum_{i \in \mathcal{R}} \frac{y_i}{\pi_i \phi_i}$$

- Where $\mathcal{R}$ is the set of units selected and who have a value of y_i recorded.

8.4 Missing Data Mechanisms

- In general, the propensity scores ϕ_i are unknown.
- How well we can estimate ϕ_i will depend on the type of missing data.
- In the next slides, I will explain missing data mechanisms both intuitively and using math notations

8.4 Missing Data Mechanisms

Some notation:

- **Observed data:** D^{obs}
- **Missing data:** D^{miss}
- $D = (D^{obs}, D^{miss})$ (complete data)
- **Missing data or nonresponse model:** $\phi = \Pr(R = 1 \mid D^{obs}, D^{miss})$

8.4 Missing Data Mechanisms

Missing completely at Random

1. Intuitive definition

- The causes of missing data are unrelated to the data.

Examples:

- Missing laboratory measurement since a test tube was randomly dropped
- Study ran out of funds
- A person's survey got lost in the mail

8.4 Missing Data Mechanisms

Missing completely at Random

2. Math definition

- $\Pr(R_i = 1 \mid D^{obs}, D^{miss}) = \Pr(R_i = 1) = \phi_i$
- The probability of nonresponse (ϕ) doesn't depend on the value of the observed or missing data
- ϕ also is unrelated to survey design.

If you take an SRS and you assume 1) $\phi_i = \phi$ for all units, 2) selection and response are independent, and 3) MCAR, then the sample mean $\bar{y}_R$ among respondents is an unbiased estimator for the population mean.

8.4 Missing Data Mechanisms

Missing completely at Random

Exercise:

- We have 3 variables that we want to include in our final model: x_1, x_2, y
- We also observe other data: z_1, z_2
- **How can I simulate MCAR mechanism in x_1 ?**

8.4 Missing Data Mechanisms

Missing at Random

1. Intuitive definition

- The probability of missing data can be explained by data that was observed

Example:

- On surveys, women are less likely to report their income, regardless of their level of income. We also know if the survey participant was female or male.
- When placed on a soft surface, a scale produces more missing values. We also know when the scale was placed on a soft surface or a hard surface.

8.4 Missing Data Mechanisms

Missing at Random

2. Math definition

- $\Pr(R = 1 \mid D^{obs}, D^{miss}) = \Pr(R = 1 \mid D^{obs})$
- The probability of nonresponse depends only on observed data
- The probability of nonresponse does NOT depend on the value of the missing data

This is our second favourite kind because we can try to fit a model to $\Pr(R = 1 \mid D^{obs})$!!

8.4 Missing Data Mechanisms

Missing at Random

Exercise:

- We have 3 variables that we want to include in our final model: x_1, x_2, y
- We also observe other data: z_1, z_2
- **How can I simulate MAR mechanism in x_1 ?**

8.4 Missing Data Mechanisms

Missing not at Random

1. Intuitive definition

- If neither MCAR or MAR holds
- Probability of nonresponse could depend on data that was unobserved and/or data that was observed.

Examples:

- Subjects with lower income are less likely to provide their income in a survey
- Laboratory values below a detection cannot be measured
- A scale produces more missing values for lighter objects

8.4 Missing Data Mechanisms

Missing not at Random

2. Math definition

- $\Pr(R = 1 \mid D^{obs}, D^{miss}) = \Pr(R = 1 \mid D^{obs}, D^{miss})$
- The probability of missing data depends on the missing data and/or data that is observed

8.4 Missing Data Mechanisms

Missing not at Random

Exercise:

- We have 3 variables that we want to include in our final model: x_1, x_2, y
- We also observe other data: z_1, z_2
- **How can I simulate MNAR mechanism in x_1 ?**

Next Class:

- Adjusting for non-response.
 - Can involve estimating ϕ_i , post-stratification, or imputation.
- Measurement error
 - Methods to minimize it.

Progress

- ☒ Distinguish between observational studies and randomized experiments and explain the types of research questions each can answer.
- ☒ Explain the concept of a causal effect and describe why randomized experiments can be used to investigate causal relationships
- ☒ Describe the missing data mechanisms MCAR, MAR, and NMAR, and explain how they differ.
- ☒ Evaluate the potential impact of nonresponse and missing data on survey estimates.
- ☒ Describe common strategies for reducing nonresponse in surveys.
- ☒ Explain the rationale behind the two-stage approach to nonresponse adjustment and describe how it can reduce nonresponse bias.
- ☒ Construct estimators under a two-stage nonresponse design